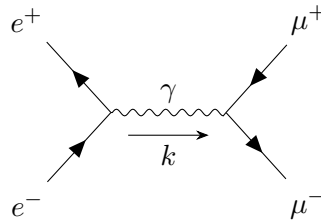


Quantum Field Theory

LECTURE NOTES

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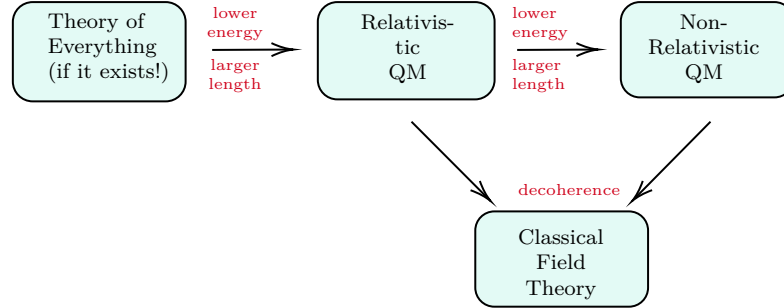
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Lecture 01: Intro and Quantum SHO

*People do 'weird' stuffs for earning,
You can do the same for learning!*

Many physical theories have been proposed and it is imperative to see how each theory is linked with the previous in some way.



From the proposed theory of everything, we can transform to a *relativistic* quantum field theory which will be the topic of discussion for most part. When subjected to *decoherence*, a relativistic QFT collapse to a classical QFT as noise hides the ‘quantumness’ from consideration. Hence we will start our discussion from classical fields and then move to quantum fields.

Since the foundational aspect of QFT (and many other topics in Physics) is the Harmonic Oscillator, let us discuss the quantum Harmonic oscillator for introduction. For that, note that the classical Hamiltonian for the harmonic oscillator is given by:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

We do not like the things like m and ω which prevent us from seeing things clearly 😊. Hence, we invoke the holy action of making things dimensionless. For that, we define:

$$\begin{aligned} X &= \sqrt{\frac{m\omega}{\hbar}} x & \longrightarrow & x^2 = \frac{\hbar}{m\omega} X^2 \\ P &= \frac{1}{\sqrt{m\omega\hbar}} p & \longrightarrow & p^2 = m\omega\hbar P^2 \end{aligned}$$

Substituting these in the Hamiltonian, we have:

$$H = \frac{\hbar\omega}{2} (X^2 + P^2)$$

Note that, we are still within the classical domain. Now, let us elevate x and p to operators and we define:

$$[x, p] = i\hbar \mathbb{1} \implies [X, P] = \sqrt{\frac{m\omega}{\hbar}} \frac{i\hbar \mathbb{1}}{\sqrt{m\omega\hbar}} = i$$

Introducing the commutator bracket brings us to the quantum world. Now, we invoke our very own ladder operators:

$$\begin{aligned} \hat{a} &= \frac{1}{\sqrt{2}}(\hat{X} + i\hat{P}) & : \text{annihilation operator} \\ \hat{a}^\dagger &= \frac{1}{\sqrt{2}}(\hat{X} - i\hat{P}) & : \text{creation operator} \end{aligned}$$

Then we will have ¹:

$$[a, a^\dagger] = \frac{1}{2} [X + iP, X - iP] = \frac{-i}{2} ([X, P] - [P, X]) = -i \times i = 1$$

¹henceforth, forsaking the hat symbol and identity operator $\mathbb{1}$, since they cause nothing but trouble, when the context is clear

Also, note that:

$$a^\dagger a = \frac{1}{2}(X^2 + P^2 + \underbrace{i(XP - PX)}_i) = \frac{1}{2}(X^2 + P^2) - \frac{1}{2} \implies H = \frac{\hbar\omega}{2}(a^\dagger a + \frac{1}{2})$$

Let us consider the *complete set of commuting observables* (CSCO) for this problem. Evidently, the set $\{H\}$ itself satisfies the condition since the eigenvalues are all non-degenerate (hence, we can label each state with only one index). To understand why, let us consider the action of the annihilation operator on a (normalised) state $|\psi\rangle$. For that we note the following:

$$\begin{aligned} [a^\dagger a, a] &= -a \implies [H, a] = -\hbar\omega a \\ [a^\dagger a, a^\dagger] &= a^\dagger \implies [H, a^\dagger] = \hbar\omega a^\dagger \end{aligned}$$

Now, we have:

$$Ha|\psi\rangle - aH|\psi\rangle = [H, a]|\psi\rangle = -\hbar\omega a|\psi\rangle \implies H(a|\psi\rangle) = (E - \hbar\omega)(a|\psi\rangle)$$

Thus, if $|\psi\rangle$ has an energy eigenvalue E , then $a|\psi\rangle$ will have an energy eigenvalue $E - \hbar\omega$. Thus, starting from any energy state, we can change to another state with energy reduced by one unit of $\hbar\omega$, using the annihilation operator. Similarly, we will have:

$$H(a^\dagger|\psi\rangle) = (E + \hbar\omega)(a^\dagger|\psi\rangle)$$

Let us denote the states $a^\dagger|\psi\rangle$ and $a|\psi\rangle$ by $|\psi_+\rangle$ and $|\psi_-\rangle$ respectively. Then, we will have

$$\langle\psi_-|\psi_-\rangle = \langle\psi|a^\dagger a|\psi\rangle = \langle\psi|\left(\frac{H}{\hbar\omega} - \frac{1}{2}\right)|\psi\rangle$$

Now, since $|\psi_-\rangle$ is a valid vector of the Hilbert space, its norm must be non-negative and finite. Hence, we obtain the condition:

$$0 \leq \frac{E}{\hbar\omega} - \frac{1}{2} < \infty \implies \frac{\hbar\omega}{2} \leq E$$

Hence, we get a lower bound on the energy eigenvalue, that is, there exists a state $|\psi_{\min}\rangle$ such that $H|\psi_{\min}\rangle = E_{\min}|\psi_{\min}\rangle$ where $E_{\min} = \frac{\hbar\omega}{2}$. Then, starting from this state, if we apply the creation operator, we will get successively increasing energies (and hence the system is non-degenerate). We thus obtain:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \quad n = 0, 1, \dots$$

Lecture 02: Dimensional Deep Dive

For the quantum harmonic oscillator, we had seen that how making things dimensionless eases calculations a bit. We consider three fundamental constants, each occupying their own special place in their field of action.

$$\begin{aligned} c &\equiv [LT^{-1}] && : \text{relativity} \\ \hbar &\equiv [L^2MT^{-1}] && : \text{quantum} \\ G &\equiv [L^3M^{-1}T^{-2}] && : \text{gravity} \end{aligned}$$

Lengths are tractable for us, since we can see the ‘length’, same goes for mass, atleast we can ‘feel’ it. However, time is an *enigma*. *We can't hold two ends of time at the same time* unlike holding two ends of a rod to measure its length. The absolute truth is: *Time passes!*

Note that, in physics, we are mostly concerned with equations like $E = mc^2$ and $E = \hbar\omega$. To that extend, we define the natural units:

$$c = 1$$

$$\hbar = 1$$

We want these quantities to be numerically equal to 1 and dimensionless. Note that, we had also done this kind of things before. When writing Newton's law, we had said that

$$F \propto m, F \propto a \implies F \propto ma \implies F = kma \quad \text{for some } k$$

Now, we chose unit and dimension of force in a way such that $k = 1$ (dimensionless and unit value) which gave us the celebrated law.

Note that:

- Making c dimensionless:

$$[LT^{-1}] \equiv 1 \implies [L] = [T]$$

- Making $\hbar$ dimensionless:

$$[L^2MT^{-1}] = [L^2ML^{-1}] = [LM] \equiv 1 \implies [L] = [M^{-1}]$$

Hence, we see that length and time are equivalent while mass and length have inverse relation. In this natural units, we have that energy is equivalent to mass and any other unit can be represented in terms of mass. Thus, by our convention, we choose mass or energy as the only important dimension.

Note that

$$\hbar c \equiv Jm = 1 \quad (\text{in natural units})$$

From this, we can say heuristically that increasing length scale is decreases the energy (mass) scale. We mainly use the length scale in context of *de Broglie wavelength*.

This is perhaps not the only way to define natural units. In cosmology, G plays a more important role and hence it is better to set $G = 1$, leaving aside $\hbar$. Using this *natural units*, we find:

$$[L] = [M]$$

$$[L] = [T]$$

Here, we see that length and mass scale are directly related. Well, in this regard, we treat the length scale to be that of the Schwarzschild radius of a blackhole ¹ which intuitively grows with mass. Since the above two natural units are widely distinct, there is as such no problem, however, in the unfortunate case where we have to consider both de Broglie wavelength and the Schwarzschild radius (that is, in the infamous domain of quantum gravity 🤖), one needs to be very careful.

Theorem 1 (π -Theorem):

Let $q_1, \dots, q_n$ be n variables which are physically relevant to a problem and which are related by an expression, that is,

$$F(q_1, \dots, q_n) = 0 \iff q_i = \tilde{F}(q_1, \dots, \hat{q}_i, \dots, q_n)$$

If k is the number of fundamental dimensions required to describe the n variables, then we can group these in $(n - k)$ groups of dimensionless variables $\Pi_1, \dots, \Pi_{n-k}$ such that for some f , we have:

$$f(\Pi_1, \dots, \Pi_{n-k}) = 0 \iff \Pi_i = \tilde{f}(\Pi_1, \dots, \hat{\Pi}_i, \dots, \Pi_{n-k})$$

.

¹This crap is the radius of an object such that if the body is squeezed to a radius lesser than the Schwarzschild radius, the gravitational attraction between the constituents of the body causes its irreversible collapse, turning it to a black hole



The theorem seems a bit vague (and pointless too). Let us take a physical example. Consider a spherical ball in a viscous fluid. The variable in the problem are:

$$\begin{aligned} \text{Drag force: } q_1 &\rightarrow F \quad [MLT^{-2}] \\ \text{Sphere diameter: } q_2 &\rightarrow d \quad [L] \\ \text{Fluid density: } q_3 &\rightarrow \rho \quad [ML^{-3}] \\ \text{Fluid velocity: } q_4 &\rightarrow v \quad [LT^{-1}] \\ \text{Fluid viscosity: } q_5 &\rightarrow \eta \quad [ML^{-1}T^{-1}] \end{aligned}$$

So, there are 5 such parameters and only three units viz. M, L, T are needed to describe them. Hence we will have two Π groups. It is a good thing to choose the repeating variables (variables which will be in both groups) that relate to mass, geometry and the kinematics of the problem. Also, note that since the Π groups are dimensionless, we can take the non-repeating variable's power to be 1. Hence, in this problem we choose them to be ρ, d, v . Thus, we will have:

$$\begin{aligned} \Pi_1 &= \rho^{a_1} d^{a_2} v^{a_3} F \equiv [ML^{-3}]^{a_1} [L]^{a_2} [LT^{-1}]^{a_3} [MLT^{-2}] = [M^{a_1+1} L^{-3a_1+a_2+a_3+1} T^{-a_3-2}] \\ \Pi_2 &= \rho^{b_1} d^{b_2} v^{b_3} \eta \equiv [ML^{-3}]^{b_1} [L]^{b_2} [LT^{-1}]^{b_3} [ML^{-1}T^{-1}] = [M^{b_1+1} L^{-3b_1+b_2+b_3-1} T^{-b_3-1}] \end{aligned}$$

Hence we obtain two sets of equations:

$$\begin{aligned} a_1 + 1 &= 0 \implies a_1 = -1 \\ -a_3 - 2 &= 0 \implies a_3 = -2 \\ -3a_1 + a_2 + a_3 + 1 &= 0 \implies 3 + a_2 - 2 + 1 = 0 \implies a_2 = -2 \\ b_1 &= -1 \\ b_3 &= -1 \\ -3b_1 + b_2 + b_3 - 1 &= 0 \implies 3 + b_2 - 2 = 0 \implies b_2 = -1 \end{aligned}$$

Then we obtain the Π groups as:

$$\Pi_1 = \frac{F}{\rho d^2 v^2} \quad \Pi_2 = \frac{\eta}{\rho d v} = \frac{1}{\frac{\rho d v}{\eta}}$$

We identify $\frac{\rho d v}{\eta}$ to be the *Reynold's numer* $\mathcal{R}$. Then we can say, for some ϕ ,

$$\frac{F}{\rho d^2 v^2} = \phi(\mathcal{R})$$

To some extraterrestrial being, the physical laws will (hopefully) be valid, however, they might not understand the human-made units (like metres and seconds) in which we measure these quantities.

We need something natural, based on Nature and hence we used the natural units. For this purpose, we will use things like $c, \hbar, G, k_b \dots$ and we set all of them to 1. Doing this will lead to change in all scales. For that, we define the Planck units, made of fundamental constants of Nature:

- **Planck mass:** $E_p = \sqrt{\frac{\hbar c}{G}}$
- **Planck length:** $l_p = \sqrt{\frac{\hbar G}{c^3}}$
- **Planck time:** $t_p = \sqrt{\frac{\hbar c}{G^5}}$

Let us now analyse the physical regimes in which the fundamental constants become important. For that, we will use a tuple $(G, \frac{1}{c}, \hbar)$ (note that all the elements of the tuple are written in terms of very small quantities of the SI scale).

$(0, 0, 0) \rightarrow$	Classical mechanics
$(G, 0, 0) \rightarrow$	Newtonian gravity
$(0, \frac{1}{c}, 0) \rightarrow$	Special relativity
$(0, 0, \hbar) \rightarrow$	Basic quantum mechanics
$(G, \frac{1}{c}, 0) \rightarrow$	General relativity
$(0, \frac{1}{c}, \hbar) \rightarrow$	QFT and relativistic QM
$(G, 0, \hbar) \rightarrow$	Non-relativistic gravity
$(G, \frac{1}{c}, \hbar) \rightarrow$	Quantum gravity

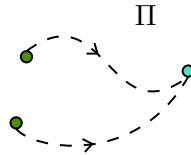
Lecture 03: From Classical to Quantum

Inspecting the transition from a classical to quantum description is necessary for understanding QFT, which is just a *sophisticated mechanics*.

We know that the position $\mathbf{q} \equiv \{q_1, \dots, q_n\}$ and momentum $\mathbf{p} \equiv \{p_1, \dots, p_n\}$ uniquely define the state of a *classical* particle.

$$(\mathbf{q}, \mathbf{p}) \in \Pi \text{ (} 2n \text{ dimensional phase space)} \quad q_i, p_i \in \mathbb{R}$$

The Hamiltonian $H(\mathbf{q}, \mathbf{p})$ is just a way of performing time evolution on the system. Using the Hamiltonian, we can ‘fibrate’ the phase space, that is, starting from one point, we can go to some other point. Also, note that, changing the parameters in the Hamiltonian (eg. ω in SHO Hamiltonian), we change the fibration patterns(eg. *Lissajou’s* figures).



In the quantum world, the phase space changes to the Hilbert space $\mathcal{H}$ while a point in the phase space (q, p) changes to a vector $|\psi\rangle$ in the Hilbert space.

Also, these real variables become Hermitian (self-adjoint) operators ¹and the classical Poisson bracket now transforms to the commutator.

$$\begin{aligned} \text{CM:} \quad & \{q, p\} = 1 \\ \text{QM:} \quad & [\hat{q}, \hat{p}] = i \end{aligned}$$

3.1 Time Evolution

From classical Hamilton’s equations of motion, we have:

$$\begin{aligned} \dot{q}_i &= \{q_i, H\} = \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= \{p_i, H\} = -\frac{\partial H}{\partial q_i} \end{aligned}$$

¹A self-adjoint operator $\mathcal{O}$ is such that $\mathcal{O} = \mathcal{O}^\dagger$ in all respect, that is, $\mathcal{O}$ and $\mathcal{O}^\dagger$ have the same domain and action. In general, $\mathbb{D}(\mathcal{O}) \subseteq \mathbb{D}(\mathcal{O}^\dagger)$ where $\mathbb{D}(\cdot)$ represents the domain of some operator. If the domains are not equal but action is same on a restricted domain (mainly occurs in infinite-dimensional spaces), then those operators are not self-adjoint but called *symmetric/Hermitian* (in some places Hermiticity also requires a symmetric operator to be bounded).

If $\mathbf{z} = \begin{pmatrix} \mathbf{p} \\ \mathbf{q} \end{pmatrix}$ is a $2n$ dimensional vector, then $\dot{\mathbf{z}} = \{\mathbf{z}, H\}$. Now, moving to quantum mechanics, we know the celebrated Schrödinger equation, which specifies the time evolution of a state vector $|\psi(t)\rangle \in \mathcal{H}$:

$$i \frac{\partial |\psi(t)\rangle}{\partial t} = H |\psi(t)\rangle$$

Thus, in both classical and quantum case, the time derivative of a quantity is equal to some action of the Hamiltonian on that quantity (classical $\rightarrow$ Poisson bracket, quantum $\rightarrow$ Multiplication with Hamiltonian)¹.

The Hamiltonian in the position basis becomes:

$$H = \frac{-\hbar^2}{2m} \nabla^2 + V(\mathbf{x})$$

Thus, the Schrödinger equation now acts on the wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi \rangle$:

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = -\frac{1}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}) \psi(\mathbf{x}, t)$$

And the energy and momentum operators become $E \rightarrow i \frac{\partial}{\partial t}$ and $p \rightarrow -i \nabla$.

3.1.1 Free Particle

The dispersion relation for a non-relativistic free particle is:

$$E = \frac{\mathbf{p}^2}{2m}$$

The typical solution can be written as plane-waves, $e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = e^{i(\mathbf{p} \cdot \mathbf{x} - Et)} \equiv \exp(-ip^\mu x_\mu)$ where we have used the repeated summation notation and taken the metric $\eta_{\mu\nu} = (+, -, -, -)$.

For a relativistic free particle, the dispersion relation becomes:

$$E^2 = m^2 + p^2$$

Instead of the wavefunction $\psi(\mathbf{x}, t)$, we will use $\phi(\mathbf{x}, t)$, since the resulting equation actually acts on a scalar-field and ϕ is common notation for a scalar-field. Substituting the energy and momentum operators here, we get:

$$\begin{aligned} i^2 \frac{\partial^2 \phi(\mathbf{x}, t)}{\partial t^2} &= (-i)^2 \nabla^2 \phi(\mathbf{x}, t) + m^2 \phi(\mathbf{x}, t) \\ \Rightarrow \frac{\partial^2 \phi(\mathbf{x}, t)}{\partial t^2} - \nabla^2 \phi(\mathbf{x}, t) &= -m^2 \phi(\mathbf{x}, t) \\ \Rightarrow \left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right) \phi(\mathbf{x}, t) &= 0 \end{aligned}$$

The above equation is called the *Klein-Gordon equation*. In covariant notation, we can compactly write it as:

$$(\partial_\mu \partial^\mu + m^2) \phi(x^\nu) = 0$$

The KG equation is Lorentz invariant, that is, under a Lorentz transformation, $x^\mu \rightarrow x'^\mu$, $\phi'(x'^\mu) = \phi(x^\mu)$

¹ A better analogy would be to use density matrices which gives the von Neumann equation which uses commutator bracket

3.1.2 Continuity Equation

Taking the complex conjugate of the Schrödinger's equation and then after some algebraic manipulation, we obtain:

$$\frac{\partial(\psi^*\psi)}{\partial t} = \frac{i}{2m} \nabla \cdot [\psi^*(\nabla\psi) - \psi(\nabla\psi^*)]$$

Here, we identify:

$$\rho := \psi^*\psi \quad \mathbf{J} = \frac{-i}{2m} [\psi^*(\nabla\psi) - \psi(\nabla\psi^*)]$$

which yields the well-known form of the *continuity equation*:

$$\frac{\partial\rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \quad \longrightarrow \quad \partial_\mu J^\mu = 0$$

where $J^\mu = (\rho, \mathbf{J})$ is the four-current. Note that in this case, ρ is a positive-definite quantity and can indeed have the interpretation of probability. Also, note that if ψ somehow becomes real-valued, then $\mathbf{J} = 0$ which implies that ρ is constant in time (though it can change in space).

Doing the same thing to Klein-Gordon equation yields:

$$\frac{\partial}{\partial t} \left(\phi^* \frac{\partial\phi}{\partial t} - \phi \frac{\partial\phi^*}{\partial t} \right) - \nabla \cdot (\phi^* \nabla\phi - \phi \nabla\phi^*) = 0$$

From this equation, we can identify:

$$\rho := \left(\phi^* \frac{\partial\phi}{\partial t} - \phi \frac{\partial\phi^*}{\partial t} \right) \quad \mathbf{J} := -(\phi^* \nabla\phi - \phi \nabla\phi^*)$$

Thus here, the 4-current becomes $J^\mu = (\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*)$. Note the apparent problems with this identification:

- It is not a priori obvious that ρ is positive-definite and hence has problem with probability interpretation.
- The equation is second-order in time and hence ρ seems to have a term evolving forward and one term evolving backward in time.
- The dispersion relation does not have a single solution; the solutions are $\pm E$

The thing is, KG equation treats time and space on equal footing (unlike Schrödinger equation where time was in first order and space was in second order). Hence we have to consider all possibilities of moving back and forth in both space and time.

Lecture 04: Classical Fields

Let us start with a familiar setting, the spring-mass system in one-dimensions. So, each mass is connected with isotropic springs of natural length a and spring constant k , and at some instance, i^{th} spring is displaced from its equilibrium position by an amount η_i as shown below:

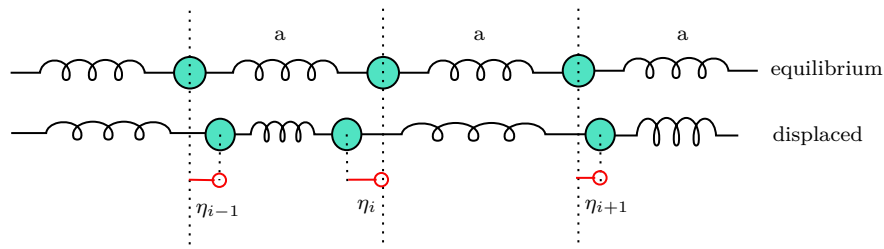


Figure 1: A spring-mass system; each mass is displaced by some amount from its equilibrium position

Then the kinetic and potential energies can be written as:

$$T = \frac{1}{2} \sum_i \dot{\eta}_i^2 \quad V = \frac{1}{2} \sum_i k(\eta_{i+1} - \eta_i)^2$$

From this, the Langrangian of the system is obtained as:

$$L = T - V = \frac{1}{2} \sum_i a \left[(m/a) \dot{\eta}_i^2 - ka \left(\frac{\eta_{i+1} - \eta_i}{a} \right)^2 \right] = a \sum_i L_i$$

The resulting ELEOM are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}_i} \right) = \frac{\partial L}{\partial \eta_i} \implies (m/a) \ddot{\eta}_i - ka \left(\frac{\eta_{i+1} - \eta_i}{a^2} \right) + ka \left(\frac{\eta_i - \eta_{i-1}}{a^2} \right) = 0$$

In the *continuum* limit, that is, $a \rightarrow 0$ (so, the system kind of becomes like a rod), we have:

$$\begin{aligned} \frac{m}{a} &\longrightarrow \mu && : \text{mass per unit length of the rod} \\ ka &\longrightarrow Y && : \text{Young's modulus of the rod} \\ i &\longrightarrow x && : \text{position label} \\ \eta_i &\longrightarrow \eta(x) && : \text{field} \\ \frac{\eta_{i+1} - \eta_i}{a} &\longrightarrow \frac{d\eta}{dx} && : \text{derivative} \\ \sum_i &\longrightarrow \int dx && : \text{integral} \end{aligned}$$

Hence in the continuum limit, the ELEOM becomes:

$$\lim_{a \rightarrow 0} \left(\mu \frac{d^2 \eta}{dt^2} - \frac{Y}{a} \frac{d\eta}{dx} \Big|_x + \frac{Y}{a} \frac{d\eta}{dx} \Big|_{x-a} \right) = 0 \implies \mu \frac{d^2 \eta}{dt^2} - Y \frac{d^2 \eta}{dx^2} = 0$$

Defining the velocity as $v := \sqrt{\frac{Y}{\mu}}$, we obtain the wave-equation:

$$\frac{1}{v^2} \frac{d^2 \eta}{dt^2} - \frac{d^2 \eta}{dx^2} = 0$$

Note that, in this case, position becomes a mere label like in the discrete case, i just labelled the index of the i^{th} mass. Also, technically, η can also depend on time, so it is better to write it as $\eta(x, t)$.

Position and time, both are kind of labels and thus are independent. Hence, total and partial derivatives coincide in this case. Note that in continuum, the Langrangian itself becomes an integral of the form:

$$L = \frac{1}{2} \int \left[\mu \dot{\eta}^2 - Y \left(\frac{d\eta}{dx} \right)^2 \right] dx$$

The bracketted term is the Lagrangian density, denoted by $\mathcal{L}$ and hence, the Lagrangian is the space integral of the Langrangian density:

$$L = \int \mathcal{L} dx$$

Henceforth, we allow ourselves to be *sloppy* enough and refer to Lagrangian density as simply the Langrangian when context is clear.

4.1 Our actions tell a lot!

Let us now consider a general setting. So, the Lagrangian density can be a function of the field, the spatial and temporal derivatives and space and time themselves. In compact notation, we write:

$$\mathcal{L} \equiv \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

Remember, that we define the classical action S as:

$$S = \int_{t_1}^{t_2} dt L = \int_{t_1}^{t_2} dt \int d^3\mathbf{x} \mathcal{L}(\phi, \partial_\mu \phi, x^\mu) = \int_{\mathcal{R} \subset \mathbb{R}^4} d^4\mathbf{x} \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

The integration can indeed be over a region $\mathcal{R}$ which is a subset of $\mathbb{R}^4$ in general. Mostly, we take $\mathcal{R} = [t_1, t_2] \times \mathbb{R}^3$.

We now want to use Hamilton's principle, which states that, under fixed boundary conditions, the system takes a path in configuration space for which the action is stationary, that is,

$$\delta S = 0$$

Note that the action is a functional which takes a field ϕ as input and returns a real number¹. Let us consider a variation in the field itself: $\phi \rightarrow \phi' = \phi + \delta\phi$ such that $\delta\phi$ vanishes at the boundary of the integration domain. Then we have the following:

$$\delta S = S[\phi'] - S[\phi] = \int d^4\mathbf{x} (\mathcal{L}(\phi', \partial_\mu \phi', x^\mu) - \mathcal{L}(\phi, \partial_\mu \phi, x^\mu))$$

Now, we try to find what the above quantity is:

$$\begin{aligned} \mathcal{L}(\phi', \partial_\mu \phi', x^\mu) &= \mathcal{L}(\phi + \delta\phi, \partial_\mu(\phi + \delta\phi), x^\mu) \\ &= \mathcal{L}(\phi, \partial_\mu \phi, x^\mu) + \frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \end{aligned}$$

Also note that $\delta(\partial_\mu \phi) = \partial_\mu(\delta\phi)$. Hence, from the above expression, we obtain that:

$$\delta S = \int d^4\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu(\delta\phi) \right)$$

Using *integration by parts* on the second term, we obtain:

$$\int d^4\mathbf{x} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu(\delta\phi) = - \int d^4\mathbf{x} \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta\phi + \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right) \Big|_{\text{boundary}}$$

The variation in the field vanishes at the boundary by our assumption. Then we obtain, for the variation of the action:

$$\delta S = \int d^4\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right) \delta\phi = 0$$

Since the variation in the field was arbitrary, we obtain the Euler-Lagrange's equations for the field as:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0$$

In general, we want the action to be *Lorentz* invariant, in order to maintain the universality of physical laws. Since the volume element of Minkowski space $d^4\mathbf{x}$ is invariant under Lorentz transformations, this imply that $\mathcal{L}$ should also be constructed of Lorentz-invariant quantities.

¹For functionals, a common notation for the input is to use square brackets, that is, if A is a linear functional taking function f as input, $A[f] \in \mathbb{R}$

The field ϕ (and all its powers) is a Lorentz *scalar* and is hence invariant. The derivative $\partial_\mu \phi$ is not invariant, however, $\partial^\mu \partial_\mu \phi$ is. Bilinears like $(\partial^\mu \phi)(\partial_\mu \phi)$ are also invariant. Hence, a simple Lagrangian should be constructed of these quantities. Let us consider the Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial^\mu \phi)(\partial_\mu \phi) - \frac{1}{2}m^2 \phi^2 = \frac{1}{2}\eta^{\mu\nu}(\partial_\nu \phi)(\partial_\mu \phi) - \frac{1}{2}m^2 \phi^2$$

Then, from the Lagrange's equations,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \phi} &= -m^2 \phi \\ \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} &= \frac{1}{2}\eta^{\mu\nu}(\partial_\nu \phi + \partial_\mu \phi \delta^\mu_\nu) \\ &= \frac{1}{2}\eta^{\mu\nu}(2 \times \partial_\nu \phi) \\ &= \eta^{\mu\nu} \partial_\nu \phi \\ &= \partial^\mu \phi\end{aligned}$$

Then we finally obtain:

$$-m^2 \phi - \partial_\mu \partial^\mu \phi = 0 \implies (\partial_\mu \partial^\mu + m^2)\phi = 0$$

This is the KG equation that we had seen earlier. Thus, the above Lagrangian models the KG equation.

When we have multiple fields $\{\phi_i\}$, then for each field, we will obtain:

$$\frac{\partial \mathcal{L}}{\partial \phi_i} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) = 0$$

NOTE: In terms of the total Lagrangian, the ELEOM for the field is:

$$\frac{\delta L}{\delta \phi} - \partial_t \left(\frac{\delta L}{\delta \dot{\phi}} \right) = 0$$

Lecture 05 & 06: Looking into Hamiltonians

Previously, we saw a ton of crap about the Lagrangians and how the equations of motions are obtained from the action being stationary. Now, we will look into the *Hamiltonian* formalism which will allow us to quantise the field.

In basic classical mechanics, we had seen that given a Lagrangian $L(q_i, \dot{q}_i, t)$, we obtained the Hamiltonian by the Legendre transformation as:

$$H(q_i, p_i, t) = p_i \dot{q}_i - L(q_i, \dot{q}_i, t)$$

where $p_i := \frac{\partial L}{\partial \dot{q}_i}$ was defined to be the generalised momenta, conjugate to the generalised coordinate q_i . Similarly, for classical fields, we can define:

$$\pi^i(x^\mu) := \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_i)} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_i}$$

This is the *canonical momentum* corresponding to the field. From the ELEOM, we obtain $\dot{\pi}^i = \frac{\delta L}{\delta \phi}$ Using this, the *Hamiltonian density* can be defined as:

$$\mathcal{H}(\phi^a, \pi_a, \partial_i \phi_a, x^\mu) := \pi^a \dot{\phi}_a - \mathcal{L}(\phi_a, \partial_\mu \phi_a, x^\mu)$$

In the final expression on the LHS, we have to replace $\dot{\phi}_a$ in terms of π^a to get the Hamiltonian density (Replacing this will get rid of all the time derivative and hence, in the Hamiltonian, we used ∂_i , the spatial derivatives only). Using this then, we get the total Hamiltonian as:

$$H[\pi^a, \phi_a] = \int d^3 \mathbf{x} \mathcal{H}(\phi_a, \pi_a, \partial_i \phi_a, x^\mu)$$

6.1 Functional Derivatives

Since action, Lagrangian, etc. are functionals, an apt way to describe their variation is through the *functional derivative*. These come under the domain of functional analysis (which is Math ) , however, a short introduction might be necessary.

Let us take a functional $\mathcal{F}[f]$ and consider its variation $\delta\mathcal{F}$ when $f \rightarrow f + \delta f$, where $\delta f = \epsilon\eta$. Here ϵ is a small parameter and η is a generic function, often called the *test function*. Now,

$$\mathcal{F}[f + \epsilon\eta] = \mathcal{F}[f] + \left. \frac{d\mathcal{F}[f + \epsilon\eta]}{d\epsilon} \right|_{\epsilon=0} \epsilon + \mathcal{O}(\epsilon^2)$$

We define the functional derivative as the first-order coefficient of the expansion, that is,

$$\frac{\delta\mathcal{F}}{\delta f} = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{F}[f + \epsilon\eta] - \mathcal{F}[f]}{\epsilon}$$

Functional derivatives seem to follow similar rules as in ordinary differential calculus (viz. linearity, Liebnitz rule, chain rule) and hence, can be treated to be operationally same as the ordinary derivative. However, exceptions are always there.

The differential (variation) of a functional is given as:

$$\delta\mathcal{F}[\phi] := \int \frac{\delta\mathcal{F}[\phi]}{\delta\phi} \delta\phi$$

Now let us try to relate functional derivative with normal derivative. For that, let us suppose that the space is discretised into cells, each of volume ΔV^i and each cell is assigned the *average value* of ϕ within the volume, that is,

$$\phi_i(t) = \frac{1}{\Delta V^i} \int_{\Delta V^i} d^3\mathbf{x} \phi(\mathbf{x}, t)$$

Then, L now depends on the discrete components, ϕ_i and $\dot{\phi}_i$ and the variation can be written as in terms of partial derivatives (since ϕ_i are now discrete):

$$\delta L = \sum_i \left(\frac{\partial L}{\partial \phi_i} \delta\phi_i + \frac{\partial L}{\partial \dot{\phi}_i} \delta\dot{\phi}_i \right) = \sum_i \frac{1}{\Delta V^i} \left(\frac{\partial L}{\partial \phi_i} \delta\phi_i + \frac{\partial L}{\partial \dot{\phi}_i} \delta\dot{\phi}_i \right) \Delta V^i$$

Now, consider the Lagrangian variation using the definition of functional derivative:

$$\delta L = \int d^3\mathbf{x} \left(\frac{\delta L}{\delta\phi} \delta\phi + \frac{\delta L}{\delta\dot{\phi}} \delta\dot{\phi} \right)$$

From the two above equations, we can make the following identifications:

$$\begin{aligned} \frac{\delta L}{\delta\phi} &= \lim_{\Delta V^i \rightarrow 0} \frac{1}{\Delta V^i} \frac{\partial L}{\partial \phi_i} \\ \frac{\delta L}{\delta\dot{\phi}} &= \lim_{\Delta V^i \rightarrow 0} \frac{1}{\Delta V^i} \frac{\partial L}{\partial \dot{\phi}_i} \end{aligned}$$

In the LHS, while taking functional derivative, $\mathbf{x}$ belongs to the i^{th} cell. Thus, we see that the functional derivative is proportional to the partial derivative. Now, given a Lagrangian, we can write the variation under $\phi \rightarrow \phi + \delta\phi$ using the Lagrangian density:

$$\delta L = \int d^3\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial (\partial_i \phi)} \delta(\partial_i \phi) + \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \delta\dot{\phi} \right)$$

Integration by parts and ignoring boundaries on the second term, we obtain:

$$\delta L = \int d^3\mathbf{x} \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_i \left(\frac{\partial \mathcal{L}}{\partial (\partial_i \phi)} \right) \right) \delta\phi + \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \delta\dot{\phi} \right)$$

Comparing this with the definition of functional derivative, we obtain:

$$\frac{\delta L}{\delta\phi} = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_i \left(\frac{\partial \mathcal{L}}{\partial (\partial_i \phi)} \right) \quad \frac{\delta L}{\delta\dot{\phi}} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

6.2 Hamilton's Equations of Motion

To obtain HEOM, we start with the variation of the total Hamiltonian,

$$\delta H = \int d^3\mathbf{x} \delta(\pi^\alpha \dot{\phi}_\alpha - \mathcal{L}) = \int d^3\mathbf{x} (\pi^\alpha \delta \dot{\phi}_\alpha + \dot{\phi}_\alpha \delta \pi^\alpha) - \delta L$$

Now, from ELEOM:

$$\delta L = \int \frac{\delta L}{\delta \phi_\alpha} \delta \phi_\alpha + \frac{\delta L}{\delta \dot{\phi}_\alpha} \delta \dot{\phi}_\alpha = \int \partial_t \left(\frac{\delta L}{\delta \dot{\phi}_\alpha} \right) \delta \phi_\alpha + \pi^\alpha \delta \dot{\phi}_\alpha = \int \dot{\pi}^\alpha \delta \phi_\alpha + \pi^\alpha \delta \dot{\phi}_\alpha$$

Using this, the variation in Hamiltonian becomes:

$$\delta H = \int d^3\mathbf{x} (\dot{\phi}_\alpha \delta \pi^\alpha - \dot{\pi}^\alpha \delta \phi_\alpha)$$

Thus, from above, we can identify:

$$\frac{\delta H}{\delta \phi_\alpha} = -\dot{\pi}^\alpha \quad \frac{\delta H}{\delta \pi^\alpha} = \dot{\phi}_\alpha$$

Also, previously we saw the relation between functional derivative of the Lagrangian and partial derivative of the Lagrangian density. That could well be extended to the Hamiltonian and we obtain:

$$\dot{\pi}^\alpha = -\frac{\delta H}{\delta \phi_\alpha} = -\frac{\partial \mathcal{H}}{\partial \phi_\alpha} + \partial_i \left(\frac{\partial \mathcal{H}}{\partial (\partial_i \phi_\alpha)} \right) \quad \dot{\phi}_\alpha = \frac{\delta H}{\delta \pi^\alpha} = \frac{\partial \mathcal{H}}{\partial \pi^\alpha} - \partial_i \left(\frac{\partial \mathcal{H}}{\partial (\partial_i \pi^\alpha)} \right)$$

In general, the Hamiltonian density does not depend on $\partial_i \pi^\alpha$, so we have $\dot{\phi}_\alpha = \frac{\partial \mathcal{H}}{\partial \pi^\alpha}$

6.3 Poisson Bracket

In the Hamiltonian formulation of classical mechanics, we defined the Poisson bracket of two quantities A, B by:

$$\{A, B\} = \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i}$$

Similarly, for fields also, given two functionals $\mathcal{F}[\phi, \pi]$ and $\mathcal{G}[\phi, \pi]$, we define the Poisson bracket by:

$$\{\mathcal{F}, \mathcal{G}\} := \int d^3\mathbf{x} \frac{\partial \mathcal{F}}{\partial \phi_\alpha} \frac{\partial \mathcal{G}}{\partial \pi^\alpha} - \frac{\partial \mathcal{F}}{\partial \pi^\alpha} \frac{\partial \mathcal{G}}{\partial \phi_\alpha}$$

Given a function $\mathcal{F}[\phi^\alpha, \pi^\alpha, t]$, note:

$$\begin{aligned} \dot{\mathcal{F}} &= \partial_t \mathcal{F} + \int d^3\mathbf{x} \left(\frac{\delta \mathcal{F}}{\delta \phi_\alpha} \frac{\partial \phi_\alpha}{\partial t} + \frac{\delta \mathcal{F}}{\delta \pi^\alpha} \frac{\partial \pi^\alpha}{\partial t} \right) \\ &= \partial_t \mathcal{F} + \int d^3\mathbf{x} \left(\frac{\delta \mathcal{F}}{\delta \phi_\alpha} \frac{\delta H}{\delta \pi^\alpha} - \frac{\delta \mathcal{F}}{\delta \pi^\alpha} \frac{\delta H}{\delta \phi_\alpha} \right) \quad (\text{from HEOM}) \\ &= \partial_t \mathcal{F} + \{\mathcal{F}, H\} \end{aligned}$$

This specifies the *time-evolution* of the functionals using the Poisson bracket. Now, let us consider the Poisson bracket of the fields ϕ_α and π^α . However, since the Poisson bracket was defined for functionals only, there is some problem. This can be resolved by noting that any function can be written as a functional depending on itself as:

$$\phi_\alpha(\mathbf{x}, t) = \int d^3\mathbf{x}' \mathbf{x}' \phi_\alpha(\mathbf{x}', t) \delta^3(\mathbf{x} - \mathbf{x}')$$

From the definition of the functional derivative, it is easy to note that:

$$\frac{\delta \phi_\alpha(\mathbf{x}, t)}{\delta \phi_\beta(\mathbf{x}', t)} = \delta_{\alpha\beta} \delta^3(\mathbf{x} - \mathbf{x}')$$

Also, ϕ and π are *independent functionals* and hence,

$$\frac{\delta\pi(\mathbf{x}, t)}{\delta\phi(\mathbf{x}', t)} = 0 \quad \frac{\delta\phi(\mathbf{x}, t)}{\delta\pi(\mathbf{x}', t)} = 0 \quad \forall \mathbf{x}, \mathbf{x}', t$$

Now, we can compute the Poisson bracket of the fields themselves and we obtain:

$$\begin{aligned} \{\phi_\alpha(\mathbf{x}, t), \pi^\beta(\mathbf{x}', t)\} &= \int d^3 \left(\mathbf{x}'' \frac{\delta\phi_\alpha(\mathbf{x}, t)}{\delta\phi_\mu(\mathbf{x}'', t)} \frac{\delta\pi^\beta(\mathbf{x}', t)}{\delta\pi^\mu(\mathbf{x}'', t)} - 0 \right) \\ &= \int d^3 (\mathbf{x}'' \delta_{\alpha\mu} \delta_{\beta\mu} \delta^3(\mathbf{x} - \mathbf{x}'') \delta^3(\mathbf{x}' - \mathbf{x}'')) \\ &= \delta_{\alpha\beta} \delta^3(\mathbf{x} - \mathbf{x}') \end{aligned}$$

We also have the other two Poisson brackets which are trivial:

$$\{\phi_\alpha, \phi_\beta\} = 0 \quad \{\pi^\alpha, \pi^\beta\} = 0$$

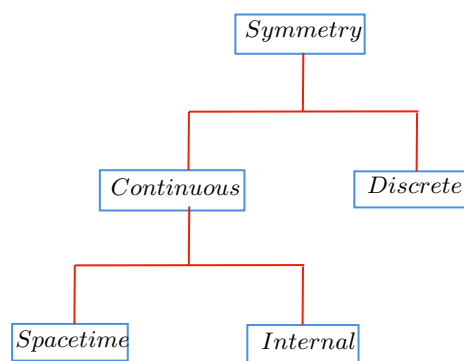
Lecture 07 & 08: Representations of Rotation and Lorentz group

A *group* is a tuple $(G, *)$ which follows certain rules like associativity, existence of identity, inverses, etc. A *Lie Group* ¹ is some continuous group with properties so special that it becomes very important (and cool) to study them.

I think a short, vague introduction cannot do justice to these elegant topics which are subjects on their own right and require proper analysis. Hence, I might skip them here and instead try to include them in the link ² below as and when I read about them!

Lecture 09: Noether's Theorem

To understand Noether's Theorem, let us first understand what symmetry actually is. Symmetry of a system can be thought of as some transformation under which the properties of the system remain invariant.



Discrete symmetries describes non-continuous changes in the system, like *parity* (kind of spatial reflection) and *time-reversal*. *Internal* symmetries include cases where the fields themselves vary while *space-time* symmetry describes the system when the space-time points vary. *Local* symmetries are transformations allowing us to change the system differently at different spacetime points while *global* symmetries act in the same way at every point.

¹A Lie Group is a group that is also a manifold.

²A very basic and nasty introduction can be found [here!](#)

Theorem 2 (Noether's Theorem):

For every continuous global symmetry there exists a conserved current J^μ , that is,

$$\partial_\mu J^\mu = 0$$

Proof. Note that, under any symmetry transformation, we required our equations of motion to be invariant and hence require our action $\mathcal{S}$ to be 'invariant' since action is a fundamental quantity to our system from which we derived our EOM. Note that while deriving the equations of motions, we ignored the boundary term of the action. Thus, the action can vary by some arbitrary boundary term. Using this, we define a *Noether symmetry* to be a transformation which keeps the action invariant upto an integral of a total divergence, that is,

$$\delta\mathcal{S} = \int_{\partial\mathcal{R}} d^4\mathbf{x} \partial_\mu K^\mu \implies \delta\mathcal{L} \equiv \epsilon \partial_\mu K^\mu$$

The implication comes from the fact that divergence should be in first order with respect to some infinitesimal variation ϵ . Let us consider the case of *internal symmetries* first.

Consider a variation in the field $\phi = \phi_0 + \delta\phi$. Then,

$$\delta\mathcal{L} = \left. \frac{\partial\mathcal{L}}{\partial\phi} \right|_{\phi_0} \delta\phi + \left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\phi_0} \delta(\partial_\mu\phi) = \left. \frac{\partial\mathcal{L}}{\partial\phi} \right|_{\phi_0} \delta\phi + \left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\phi_0} \partial_\mu(\delta\phi)$$

Now, suppose there exists a solution $\tilde{\phi}$ to the equation of motion of the system. Since the above equation is valid for any ϕ_0 , it is also valid for $\tilde{\phi}$ and hence, we get:

$$\begin{aligned} \delta\mathcal{L} &= \left. \frac{\partial\mathcal{L}}{\partial\phi} \right|_{\tilde{\phi}} \delta\phi + \left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \partial_\mu(\delta\phi) \\ &= \partial_\mu \left(\left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \delta\phi + \left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \partial_\mu(\delta\phi) \right) \quad (\text{using ELEOM}) \\ &= \partial_\mu \left(\left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \delta\phi \right) \quad (\text{using product rule}) \end{aligned}$$

Let the variation of the Lagrangian be a global symmetry, denoted by, $\delta_\alpha\phi$. Under this symmetry, we want our action $\mathcal{S}$ to change atmost by a boundary term and hence we have,

$$\partial_\mu \left(\left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \delta\phi \right) = \partial_\mu K^\mu \implies \partial_\mu \left(\left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \delta\phi - K^\mu \right) = 0$$

Let us define the quantity in the bracket by J^μ which is called the *Noether current*. Thus,

$$J^\mu := \left. \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right|_{\tilde{\phi}} \delta\phi - K^\mu$$

from where we get $\partial_\mu J^\mu = 0 \implies$ Noether current is conserved.

The above proof worked for internal symmetries where the field itself was varied. Now, let us look into the *spacetime* symmetry, where the spacetime points are varied. For simplicity, we will deal with scalar fields only. Let us consider an infinitesimal coordinate transformation,

$$x'^\mu = x^\mu + \delta x^\mu$$

The variation in the field can be given by:

$$\delta\phi = \phi'(\mathbf{x}') - \phi(\mathbf{x}) = \phi'(x'^\mu) - \phi'(x^\mu) + \phi'(x^\mu) - \phi(x^\mu) = \delta_0\phi(x^\mu) + \phi'(x'^\mu) - \phi'(x^\mu)$$

where we have defined $\delta_0\phi(x^\mu) := \phi'(x^\mu) - \phi(x^\mu)$. We did this to distinguish between variation of the dependent and independent quantities. Using Taylor expansion, we have:

$$\phi'(x'^\mu) = \phi'(x^\mu + \delta x^\mu) = \phi'(x^\mu) + \frac{\partial\phi'}{\partial x^\mu}\delta x^\mu \implies \phi'(x'^\mu) - \phi'(x^\mu) = \partial_\mu\phi'\delta x^\mu$$

Now, $\partial_\mu\phi'\delta x^\mu \approx \partial_\mu\phi\delta x^\mu$, since if we do the Taylor expansion of $\delta_0\phi(x'^\mu)$, we would obtain a term $\partial_\nu\delta_0$ which is a variation of second order and hence, can be neglected. Thus, we obtain:

$$\phi'(x'^\mu) - \phi'(x^\mu) = \partial_\mu\phi\delta x^\mu \implies \delta\phi = \delta_0\phi(x^\mu) + \partial_\mu\phi\delta x^\mu$$

The action is not in general invariant under both changes in coordinates and fields. Indeed, the volume element (integral measure) $d^4\mathbf{x}$ transforms as:

$$d^4\mathbf{x}' = J d^4\mathbf{x}$$

where J is the determinant of the Jacobian matrix. Now, we find some expression for the Jacobian:

$$x'^\mu = x^\mu + \delta x^\mu \implies \partial_\nu x'^\mu = \delta^\mu_\nu + \partial_\nu(\delta x^\mu)$$

Now, we use an identity $\det(\mathbb{1} + \epsilon A) = 1 + \epsilon \text{Tr } A + \mathcal{O}(\epsilon^2)$ on the above equation to get an approximation for the Jacobian as:

$$J = \det\left(\frac{\partial x'^\mu}{\partial x^\nu}\right) = \det(\delta^\mu_\nu + \partial_\nu(\delta x^\mu)) \approx (1 + \partial_\nu(\delta x^\mu)) \implies \delta J = \partial_\mu(\delta x^\mu)$$

Now, we consider the variation of the action itself and we see:

$$\begin{aligned} \delta\mathcal{S} &= \int \delta(d^4\mathbf{x}) \mathcal{L} + \int d^4\mathbf{x} \delta\mathcal{L} \\ &= \int d^4\mathbf{x} \partial_\mu\delta x^\mu \mathcal{L} + \int d^4\mathbf{x} \delta\mathcal{L} \\ &= \int d^4\mathbf{x} \left(\partial_\mu\delta x^\mu \mathcal{L} + \frac{\partial\mathcal{L}}{\partial\phi}\delta_0\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0(\partial_\mu\phi) + \frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu \right) \end{aligned}$$

We had written the variation of the Lagrangian with respect to both dependent and independent variables. Now the first and last term can be written together using product rule and hence we have:

$$\begin{aligned} \delta\mathcal{S} &= \int d^4\mathbf{x} \left(\partial_\mu(\delta x^\mu \mathcal{L}) + \frac{\partial\mathcal{L}}{\partial\phi}\delta_0\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\delta_0\phi \right) \\ &= \int d^4\mathbf{x} \left(\partial_\mu(\delta x^\mu \mathcal{L}) + \frac{\partial\mathcal{L}}{\partial\phi}\delta_0\phi + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0\phi\right) - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta_0\phi \right) \\ &= \int d^4\mathbf{x} \left(\partial_\mu\left(\delta x^\mu \mathcal{L} + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0\phi\right)\right) + \delta_0\phi\left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\right) \right) \end{aligned}$$

For our field satisfying the equation of motion, the last term drops and we have our variation of the action and this should be equal to the boundary term, hence we have:

$$\partial_\mu\left(\delta x^\mu \mathcal{L} + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta_0\phi\right)\right) = \partial_\mu K^\mu$$

In the case for multiple fields, the *Noether current* turns out to be:

$$J^\mu := \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\delta_0\phi_i + \delta x^\mu \mathcal{L} - K^\mu \quad \partial_\mu J^\mu = 0 \quad \square$$

The above condition is called *current conservation* which we had also seen earlier for internal symmetries. However, the second term, which is purely an artifact of the spacetime symmetry, did not occur there in the conserved current for internal symmetries. Associated with each current, we can define the charge enclosed in a volume V as:

$$Q_V := \int_V d^3\mathbf{x} J^0(\mathbf{x}, t) = \int_V d^3\mathbf{x} \rho$$

Then, we can write the conservation equation as:

$$\frac{d\rho}{dt} + \nabla \cdot \mathbf{J} = 0 \implies \frac{dQ}{dt} = - \int_V d^3\mathbf{x} \nabla \cdot \mathbf{J} = - \int_{\partial V} \mathbf{J} \cdot d\mathbf{S}$$

The last equality comes from Gauss' law and ∂V is the boundary of the region. If we extend the volume V to cover the entire space and assume that at spatial infinities, current is zero, then we have $\frac{dQ}{dt} = 0$ and hence, current is conserved. In most cases, we would have K^μ to be zero (and thus, let us take it to be zero). Let us see some examples for Noether's theorem:

9.1 Examples: Spacetime Symmetries

Let us start with some examples of Noether's theorem where spacetime symmetry is explicitly there. We will consider translations and rotations.

9.1.1 Translational Invariance

Classically, we know that translational invariance leads to *energy conservation*. Let us see what happens in case of fields. We thus consider an infinitesimal global translation:

$$x^\mu \rightarrow x'^\mu = x^\mu + \epsilon a^\mu$$

Then $\delta x^\mu = \epsilon a^\mu$. As calculated before, the variation in the Lagrangian is given by:

$$\delta \mathcal{L} = \partial_\mu \left(\delta x^\mu \mathcal{L} + \delta_0 \phi_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \right)$$

Note that $\delta x^\mu = \epsilon a^\mu$ and $\delta_0 \phi_i = \phi'_i(x^\mu) - \phi_i(x^\mu)$. Note that the new field at x^μ is just the previous field at $x^\mu - \epsilon a^\mu$ and hence we have from a first-order expansion that $\delta_0 \phi_i = -\epsilon a^\nu \partial_\nu \phi_i$. Substituting this above, we obtain:

$$\begin{aligned} \partial_\mu \left(\delta x^\mu \mathcal{L} + \delta_0 \phi_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \right) &= \partial_\mu \left(\epsilon a^\mu \mathcal{L} - \epsilon a^\nu \partial_\nu \phi_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \right) \\ &= \partial_\mu \left(\epsilon a^\nu \delta^\mu_\nu \mathcal{L} - \epsilon a^\nu \partial_\nu \phi_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \right) \\ &= -\epsilon a^\nu \partial_\mu \left(\partial_\nu \phi_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} - \delta^\mu_\nu \mathcal{L} \right) \end{aligned}$$

From the above, we can thus define the conserved current as:

$$T^\mu_\nu := \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \partial_\nu \phi_i - \delta^\mu_\nu \mathcal{L}$$

Now, note the following:

$$T^{\mu\nu} = g^{\alpha\nu} T^\mu_\alpha = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} g^{\alpha\nu} \partial_\alpha \phi_i - g^{\alpha\nu} \delta^\mu_\alpha \mathcal{L} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \partial^\nu \phi_i - g^{\mu\nu} \mathcal{L}$$

The above quantity can be shown to be a tensor and is called the *energy-momentum stress tensor*. Why?

Note that for each value of ν we have a current four vector. Let us calculate the charge for the time-translation invariance part, that is, $\nu = 0$. From this, we have:

$$P^0 = \int d^3\mathbf{x} T^{00} = \int d^3\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_i)} \partial^0 \phi_i - g^{00} \mathcal{L} \right) = \int d^3\mathbf{x} (\pi^i \dot{\phi}_i - \mathcal{L}) = \int d^3\mathbf{x} \mathcal{H} = H$$

We see that corresponding to the conserved current from time translation invariance, we have a conserved charge $P^0 = H$. Thus, the total Hamiltonian is conserved under time translation and hence our classical intuition worked here as well!

Now, let us see what happens for space translations, for which we consider $\nu \equiv j = 1, 2, 3$. Then we have:

$$P^j = \int d^3\mathbf{x} T^{0j} = \int d^3\mathbf{x} \left(\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_i)} \partial^j \phi_i - g^{0j} \mathcal{L} \right) = \int d^3\mathbf{x} (\pi^i \partial^j \phi_i)$$

The last thing is the j^{th} component of the total momentum which is conserved in this case. Thus, we see that for spatial translation invariance, the total momentum is conserved, which once again matches with the classical analog.

9.1.2 Lorentz Invariance

An infinitesimal Lorentz transformation (which covers both *boosts* and *rotations*) can be written as:

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$$

where $\omega^\mu{}_\nu$ is an infinitesimal deviation from the identity. Now, we know for any Lorentz transformation,

$$\Lambda^\mu{}_\tau \eta^{\tau\sigma} \Lambda^\nu{}_\sigma = \eta^{\mu\nu} \iff \Lambda^T \eta \Lambda = \eta$$

If we put the infinitesimal transformation in the above equation and retain only terms upto first order in ω , we would have $\omega^{\mu\nu} = \omega^{\nu\mu}$, that is, the infinitesimal deviation is an anti-symmetric matrix. The spacetime coordinates transforms as:

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu = \delta^\mu{}_\nu x^\nu + \omega^\mu{}_\nu x^\nu = x^\mu + \omega^\mu{}_\nu x^\nu \implies \delta x^\mu = \omega^\mu{}_\nu x^\nu$$

And the variation in ϕ is given as:

$$\delta_0 \phi_i = \phi'(x^\mu) - \phi(x^\mu) = \phi(x^\mu + \omega^\mu{}_\nu x^\nu) - \phi(x^\mu) = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi_i$$

Now, we are ready to put in the variation of the Lagrangian:

$$\begin{aligned} \partial_\mu \left(\delta x^\mu \mathcal{L} + \delta_0 \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) &= \partial_\mu \left(\omega^\mu{}_\nu x^\nu \mathcal{L} - \omega^\sigma{}_\nu x^\nu \partial_\sigma \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) \\ &= \partial_\mu \left(\eta^{\mu\rho} \omega_{\rho\nu} x^\nu \mathcal{L} - \eta^{\sigma\lambda} \omega_{\lambda\nu} x^\nu \partial_\sigma \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) \\ &= \partial_\mu \left(\eta^{\mu\rho} \omega_{\rho\nu} x^\nu \mathcal{L} - \omega_{\lambda\nu} x^\nu \partial^\lambda \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) \\ &= \partial_\mu \left(\eta^{\mu\rho} \omega_{\rho\nu} x^\nu \mathcal{L} - \omega_{\rho\nu} x^\nu \partial^\rho \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) \\ &= \partial_\mu \left(-\omega_{\rho\nu} x^\nu \times \left(-\eta^{\mu\rho} \mathcal{L} + \partial^\rho \phi_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) \right) \\ &= \partial_\mu (-\omega_{\rho\nu} x^\nu T^{\rho\mu}) \\ &= \partial_\mu (\omega_{\nu\rho} x^\nu T^{\rho\mu}) \end{aligned}$$

Since the current must be conserved for all the six elements of the anti-symmetric matrix (since anti-symmetric so there are total $\frac{4 \times (4-1)}{2} = 6$ independent entries), the part of $x^\nu T^{\rho\mu}$ which is anti-symmetric in ρ and μ must be conserved. After antisymmetrising and considering current for each component, we have:

$$M^{\nu\mu\rho} := x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu} \quad \partial_\mu M^{\nu\mu\rho} = 0$$

Let us now see the conserved charges for this case. Corresponding to spatial rotations, we have:

$$\mathcal{J}^{ij} = \int d^3\mathbf{x} M^{i0j} = \int d^3\mathbf{x} (x^i T^{0j} - x^j T^{0i})$$

This can be related to the angular-momentum since T^{0i} was related to P^i , the i^{th} component of the momentum. Hence, rotational invariance leads to the *conservation of angular momentum*. Let us see what happens when we give *boosts*.

$$\mathcal{J}^{0j} = \int d^3\mathbf{x} M^{00j} = \int d^3\mathbf{x} (x^0 T^{0j} - x^j T^{00})$$

As $t = x^0$ and we know that the current $\mathcal{J}^{0j}$ must be conserved, we have:

$$\frac{d\mathcal{J}^{0j}}{dt} = \frac{d}{dt} \int d^3\mathbf{x} (x^0 T^{0j} - x^j T^{00}) = \int d^3\mathbf{x} T^{0j} + \int d^3\mathbf{x} t \partial_t T^{0j} - \frac{d}{dt} \int d^3\mathbf{x} x^j T^{00} = 0$$

In terms of P^j , the above equation can be written as:

$$P^j + t \frac{dP^j}{dt} - \frac{d}{dt} \int d^3\mathbf{x} x^j T^{00} = 0$$

Previously we had seen that P^j was the conserved charge corresponding to spatial translation and hence P^j is a constant, which gives us:

$$\frac{d}{dt} \int d^3\mathbf{x} x^j T^{00} = \mathcal{C} \text{ (const.)}$$

The above relation can be seen as the centre of energy (or mass) of the field travels with constant velocity as T^{00} can be treated as the Hamiltonian ($\approx$ energy) density.

9.2 Examples: Internal Symmetries

Let us now look at internal symmetries, where the spacetime coordinates themselves do not change but the fields change due to some global transformations.

9.2.1 Global U(1) Symmetry of Complex Scalar Field

Consider a complex scalar field ϕ with the following Lagrangian:

$$\mathcal{L} = (\partial_\mu \phi)(\partial^\mu \phi^*) - m^2 \phi \phi^*$$

Let us consider a $U(1)$ transformation¹. Note that elements of the group $U(1)$ are represented by $e^{i\alpha}$, that is, they lie on the unit circle. Then, for an infinitesimal change $\psi \rightarrow e^{i\alpha}\psi$, $e^{i\alpha} = 1 + i\alpha + \mathcal{O}(\alpha^2)$ ² and we have:

$$\begin{aligned} \phi' &= (1 + i\alpha)\phi \implies \delta_\alpha \phi = i\alpha \\ \phi^{*'} &= (1 - i\alpha)\phi^* \implies \delta_\alpha \phi^* = -i\alpha \end{aligned}$$

Under this transformation, we see that the Lagrangian is invariant, that is,

$$\mathcal{L}' = (\partial_\mu \phi')(\partial^\mu \phi'^*) - m^2 \phi' \phi'^* = e^{i\alpha} e^{-i\alpha} (\partial_\mu \phi)(\partial^\mu \phi^*) - e^{i\alpha} e^{-i\alpha} m^2 \phi \phi^* = \mathcal{L}$$

From this, we can take $K^\mu = 0$. Using the formula for the conserved current, we have:

$$J^\mu = i\alpha \phi (\partial^\mu \phi^*) - i\alpha \phi^* (\partial^\mu \phi) \sim i(\partial^\mu \phi^* \phi - \partial^\mu \phi \phi^*)$$

The conserved charge is then given by the volume integral:

$$Q = \int d^3\mathbf{x} J^0 = -i \int d^3\mathbf{x} (\phi \dot{\phi}^* - \dot{\phi} \phi^*)$$

¹This is an *internal* symmetry as spacetime points are not varied.

²This is a *global* symmetry as α does not depend on the spacetime coordinates

Lecture 10 & 11: Field Quantisation

So far we had worked with classical fields. In the introduction, we saw that, quantum theories due to decoherence, transforms to classical theories, with an inevitable loss of information. Thus, finding the quantum theory leading to the classical model (working backwards) is not *unique*. More than one quantum field theory might have the same classical limit. Thus, while doing the quantisation process, we always have to make an *educated guess*!

Now, let us make the KG field *quantum* by quantising the canonical coordinates, similar to as we did for the harmonic oscillator.

We have our classical field $\phi(\mathbf{x}, t)$ and its canonical conjugate momentum $\pi(\mathbf{x}, t)$ which become operators $\hat{\phi}(x)$ and $\hat{\pi}(x)$ ¹. The classical Poisson bracket changes to the commutator, that is, $\{\cdot, \cdot\} \rightarrow i[\cdot, \cdot]$.

A quantum treatment of a classical Hamiltonian is sometimes messy since we have to maintain the Hermiticity condition and there can be multiple ways to do that. However, almost always, we just put the hats on appropriate places and wish we are correct!

Classically, we had $\{\phi(\mathbf{x}), \pi(\mathbf{y})\} = \delta^{(3)}(\mathbf{x} - \mathbf{y})$ and by the same logic:

$$[\hat{\phi}(\mathbf{x}), \hat{\pi}(\mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y})$$

Let us take their Fourier transforms,

$$\begin{aligned}\tilde{\phi}(\mathbf{k}, t) &= \int d^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}} \phi(\mathbf{x}, t) \\ \tilde{\pi}(\mathbf{k}, t) &= \int d^3\mathbf{x} e^{-i\mathbf{k}\cdot\mathbf{x}} \pi(\mathbf{x}, t)\end{aligned}$$

Let us now consider the Klein-Gordon equation,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 \iff (\partial_\mu \partial^\mu + m^2)\phi = 0$$

Expanding the field in Fourier space, we would obtain the following:

$$\left(\frac{\partial^2}{\partial t^2} + (|\mathbf{p}|^2 + m^2)\right) \tilde{\phi}(\mathbf{p}, t) = 0$$

Note that this is the equation for a classical harmonic oscillator with frequency $\omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$ and the general solution $\tilde{\phi}(\mathbf{x}, t)$ will be a linear combination of $e^{i\omega_{\mathbf{p}}t}$ and $e^{-i\omega_{\mathbf{p}}t}$. Thus we have,

$$\tilde{\phi}(\mathbf{p}, t) = A(\mathbf{p})e^{i\omega_{\mathbf{p}}t} + B(\mathbf{p})e^{-i\omega_{\mathbf{p}}t} \quad \omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$$

Now, in the quantum domain, $A(\mathbf{p})$ and $B(\mathbf{p})$ will become the creation and annihilation operators $a_{\mathbf{p}}$ and $a_{\mathbf{p}}^\dagger$ and in that spirit, let us define the field operators in terms of the annihilation and creation operators for each Fourier mode. Note that, for simplicity we define the field operators at $t = 0$.

$$\begin{aligned}\hat{\phi}(\mathbf{x}) &:= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} + \hat{a}_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}}) \\ \hat{\pi}(\mathbf{x}) &:= \int \frac{d^3\mathbf{p}}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\mathbf{p}}}{2}} (\hat{a}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} - \hat{a}_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}})\end{aligned}$$

where $\omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$ is the frequency of the mode with momentum $\mathbf{p}$. Replacing $\mathbf{p}$ with $-\mathbf{p}$ in the second term, the above can also be written in a more compact way as:

$$\begin{aligned}\hat{\phi}(\mathbf{x}) &:= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger) e^{i\mathbf{p}\cdot\mathbf{x}} \\ \hat{\pi}(\mathbf{x}) &:= \int \frac{d^3\mathbf{p}}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\mathbf{p}}}{2}} (\hat{a}_{\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger) e^{i\mathbf{p}\cdot\mathbf{x}}\end{aligned}$$

¹Note that we will write four vectors with non-bold letter, as x will denote the four components of the four-vector. To denote the three vectors, we will write in bold-face, that is, $\mathbf{x}$

Let us check the commutation relation:

$$[\hat{\phi}(\mathbf{x}), \hat{\pi}(\mathbf{y})] = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{(-i)}{2} \sqrt{\frac{\omega_{\mathbf{p}}}{\omega_{\mathbf{p}'}}} ([\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger, \hat{a}_{\mathbf{p}'} - \hat{a}_{-\mathbf{p}'}^\dagger]) e^{i(\mathbf{p}\cdot\mathbf{x} + \mathbf{p}'\cdot\mathbf{y})} = i\delta^{(3)}(\mathbf{x} - \mathbf{y})$$

Note that the above relation can be satisfied if we take:

$$[\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \mathbb{1}$$

Now, let us do a horrendous calculation, which will then lead to a horrendous result and after some reasoning based on physical grounds, we will obtain a cute result. From the Hamiltonian density, we have the total Hamiltonian as:

$$\hat{H} = \int d^3\mathbf{x} \mathcal{H} = \int d^3\mathbf{x} \left(\frac{1}{2} \hat{\pi}^2 + \frac{1}{2} (\nabla \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 \right)$$

Let us calculate this term by term. The first term gives us:

$$\begin{aligned} \int d^3\mathbf{x} \pi^2(\mathbf{x}) &= \int d^3\mathbf{x} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}}{-2} (\hat{a}_{\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{\mathbf{p}'} - \hat{a}_{-\mathbf{p}'}^\dagger) e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}}{-2} (\hat{a}_{\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{\mathbf{p}'} - \hat{a}_{-\mathbf{p}'}^\dagger) \delta^{(3)}(\mathbf{p} + \mathbf{p}') \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left[-\frac{\omega_{\mathbf{p}}}{2} \right] (\hat{a}_{\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{-\mathbf{p}} - \hat{a}_{\mathbf{p}}^\dagger) \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left[-\frac{\omega_{\mathbf{p}}}{2} \right] (\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} - \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^\dagger - \hat{a}_{-\mathbf{p}}^\dagger\hat{a}_{-\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}}^\dagger) \end{aligned}$$

From the last term, we have:

$$\begin{aligned} \int d^3\mathbf{x} \phi^2(\mathbf{x}) &= \int d^3\mathbf{x} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}} (\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{\mathbf{p}'} + \hat{a}_{-\mathbf{p}'}^\dagger) e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}} (\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{\mathbf{p}'} + \hat{a}_{-\mathbf{p}'}^\dagger) \delta^{(3)}(\mathbf{p} + \mathbf{p}') \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left[\frac{1}{2\omega_{\mathbf{p}}} \right] (\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}^\dagger) \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left[\frac{1}{2\omega_{\mathbf{p}}} \right] (\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^\dagger + \hat{a}_{-\mathbf{p}}^\dagger\hat{a}_{-\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}}^\dagger) \end{aligned}$$

Now comes the second term. For this, note that we have $\nabla(e^{i\mathbf{k}\cdot\mathbf{x}}) = i\mathbf{k}(e^{i\mathbf{k}\cdot\mathbf{x}})$. Using this, we obtain:

$$\begin{aligned} \int d^3\mathbf{x} (\nabla \phi) \cdot (\nabla \phi) &= \int d^3\mathbf{x} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}} [i\mathbf{p}(\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)e^{i\mathbf{p}\cdot\mathbf{x}}] \cdot [i\mathbf{p}'(\hat{a}_{\mathbf{p}'} + \hat{a}_{-\mathbf{p}'}^\dagger)e^{i\mathbf{p}'\cdot\mathbf{x}}] \\ &= - \int d^3\mathbf{x} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}} [(\mathbf{p} \cdot \mathbf{p}')(\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{\mathbf{p}'} + \hat{a}_{-\mathbf{p}'}^\dagger)] e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\ &= - \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}} [(\mathbf{p} \cdot \mathbf{p}')(\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{\mathbf{p}'} + \hat{a}_{-\mathbf{p}'}^\dagger)] \delta^{(3)}(\mathbf{p} + \mathbf{p}') \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} [(\mathbf{p}^2)(\hat{a}_{\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger)(\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}^\dagger)] \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\mathbf{p}^2}{2\omega_{\mathbf{p}}} (\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^\dagger + \hat{a}_{-\mathbf{p}}^\dagger\hat{a}_{-\mathbf{p}} + \hat{a}_{-\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}}^\dagger) \end{aligned}$$

Adding all the terms with correct pre-factors, we obtain the following:

$$\begin{aligned}
\hat{H} &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left\{ \left[-\frac{\omega_{\mathbf{p}}}{4} \right] (\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} - \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} - \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}) \right\} + \left\{ \left[\frac{m^2}{4\omega_{\mathbf{p}}} \right] (\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}) \right\} \\
&\quad + \left\{ \frac{\omega_{\mathbf{p}}^2 - m^2}{4\omega_{\mathbf{p}}} (\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}) \right\} \\
&= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\omega_{\mathbf{p}}}{4} (\cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}} + \cancel{\hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}} - \cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}} - \cancel{\hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}}) \\
&\quad + \frac{m^2}{4\omega_{\mathbf{p}}} (\cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} + \cancel{\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger}} + \cancel{\hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}}} + \cancel{\hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}} - \cancel{\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}} - \cancel{\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger}} - \cancel{\hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}}} - \cancel{\hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}^{\dagger}}) \\
&= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\omega_{\mathbf{p}}}{2} (\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} + \hat{a}_{-\mathbf{p}}^{\dagger}\hat{a}_{-\mathbf{p}}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\omega_{\mathbf{p}}}{2} (\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^{\dagger} + \hat{a}_{\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}})
\end{aligned}$$

Now, using the commutator as obtained before, we get:

$$\hat{H} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\omega_{\mathbf{p}}}{2} ((2\pi)^3 \delta^{(3)}(0) + 2\hat{a}_{\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \omega_{\mathbf{p}} \hat{a}_{\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}} + \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{\omega_{\mathbf{p}}}{2} \delta^{(3)}(0)$$

This is the end of the horrendous calculation and now we discuss the horrendous term (the second term) which was obtained as a result of applying the commutator. Since this contains a delta-function (distribution?) and the integral is over all momentum, this definitely cannot be a finite quantity, hence it diverges. This term basically represents the zero-point energies of the harmonic oscillators (since there are infinite number of harmonic oscillators, we have a divergent zero-point energy). The fortunate thing is that this is a constant.

When physically measuring energies, only differences can be calculated and hence this term gets cancelled and thus, no divergence actually happens.¹

Thus, in all practicality, we can rescale our energy and we obtain the Hamiltonian as:

$$\hat{H} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \omega_{\mathbf{p}} \hat{a}_{\mathbf{p}}^{\dagger}\hat{a}_{\mathbf{p}}$$

Now, let us see the commutator of the ladder operators with the Hamiltonian. We have:

$$[\hat{H}, \hat{a}_{\mathbf{p}}^{\dagger}] = \int \frac{d^3\mathbf{p}'}{(2\pi)^3} \omega_{\mathbf{p}'} [\hat{a}_{\mathbf{p}'}^{\dagger}\hat{a}_{\mathbf{p}'}^{\dagger}, \hat{a}_{\mathbf{p}}^{\dagger}] = \int \frac{d^3\mathbf{p}'}{(2\pi)^3} \omega_{\mathbf{p}'} \hat{a}_{\mathbf{p}'}^{\dagger} [\hat{a}_{\mathbf{p}'}^{\dagger}, \hat{a}_{\mathbf{p}}^{\dagger}] = \int \frac{d^3\mathbf{p}'}{(2\pi)^3} \omega_{\mathbf{p}'} \hat{a}_{\mathbf{p}'}^{\dagger} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') = \omega_{\mathbf{p}} \hat{a}_{\mathbf{p}}^{\dagger} \quad (1)$$

$$[\hat{H}, \hat{a}_{\mathbf{p}}] = \int \frac{d^3\mathbf{p}'}{(2\pi)^3} \omega_{\mathbf{p}'} [\hat{a}_{\mathbf{p}'}^{\dagger}\hat{a}_{\mathbf{p}'}^{\dagger}, \hat{a}_{\mathbf{p}}] = \int \frac{d^3\mathbf{p}'}{(2\pi)^3} \omega_{\mathbf{p}'} [\hat{a}_{\mathbf{p}'}^{\dagger}, \hat{a}_{\mathbf{p}}] \hat{a}_{\mathbf{p}'} = - \int \frac{d^3\mathbf{p}'}{(2\pi)^3} \omega_{\mathbf{p}'} \hat{a}_{\mathbf{p}'} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') = -\omega_{\mathbf{p}} \hat{a}_{\mathbf{p}} \quad (2)$$

11.1 Conserved Charges

Note that we already found the Hamiltonian which is one of the conserved charges corresponding to T^{00} . Let us now focus on the momentum, that is, $\hat{P}^j$ corresponding to T^{0j}

$$\hat{P}^j = \int d^3\mathbf{x} \hat{T}^{0j} = - \int d^3\mathbf{x} \hat{\pi} \partial_j \hat{\phi}$$

Note that, while changing from the classical expression to quantum by ‘putting hats’, we had several choices for the ordering of $\hat{\pi}$ and $\hat{\phi}$ but here, we specify the one which works!

We now have two options, either to do the calculation again or just specify the result of this calculation

¹This infinite term is sometimes attributed to the *cosmological constant*. It also has some role in the *Casimir effect*.

since this had already been done by stalwarts before. We choose the former and thus, we have:

$$\begin{aligned} \int d^3\mathbf{x} \hat{\pi} \partial_j \hat{\phi} &= \int d^3\mathbf{x} \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\mathbf{p}}}{2}} \frac{1}{\sqrt{2\omega_{\mathbf{p}'}}} (\hat{a}_{\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger) i p'_j (\hat{a}_{\mathbf{p}'} + \hat{a}_{-\mathbf{p}'}^\dagger) e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{p}}}{2}} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger) p_j (\hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}}^\dagger) \\ &= \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} p_j (\hat{a}_{\mathbf{p}} \hat{a}_{-\mathbf{p}} + \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger - \hat{a}_{-\mathbf{p}}^\dagger \hat{a}_{-\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}}^\dagger) \end{aligned}$$

Now, note the following:

$$\mathcal{I} = \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} p_j (\hat{a}_{\mathbf{p}} \hat{a}_{-\mathbf{p}} - \hat{a}_{-\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}}^\dagger) \quad \mathbf{p} \rightarrow -\mathbf{p} \quad \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} (-p_j) (\hat{a}_{-\mathbf{p}} \hat{a}_{\mathbf{p}} - \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{-\mathbf{p}}^\dagger) = -\mathcal{I}$$

The last equality comes since the creation and annihilation operators commute for different values of the momentum. Thus, this integral is zero and the expression for the total momentum becomes:

$$\hat{P}^j = \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} p_j (\hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger - \hat{a}_{-\mathbf{p}}^\dagger \hat{a}_{-\mathbf{p}}) = \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} p_j (\hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger + \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}}) = \frac{1}{2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} p_j \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}}$$

In the last term, we introduced the commutator and removed the constant infinity term.

11.2 Normal Ordering

The way we are doing, like removing the infinities, is a common thing in QFT. This purely results from the fact the things commute in the classical case and hence there is always an order ambiguity when we move to quantum.

For example, if in classical scenario, we have some term like xp , when elevating (idk if its really elevating) these things to operators, we can have multiple possibilities, viz. $\hat{x}\hat{p}$, $\hat{p}\hat{x}$ or even $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ (placing things symmetrically). Each one would present some additional constant on the final result, since the commutator of $\hat{p}$ and $\hat{x}$ is just a constant.

To be consistent, we generally use *normal ordering*, that is, keeping all the creation operators to the left. For that we define the *normal ordering operator*, $\hat{\mathcal{N}}$. For example,

$$\hat{\mathcal{N}}[\hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}}] = \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}} \quad \hat{\mathcal{N}}[\hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger] = \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger \quad \hat{\mathcal{N}}[\hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger] = \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}} \quad \dots$$

The $\hat{\mathcal{N}}$ is not an operator in the sense that it doesn't act on a state to provide some new information. Instead, normal ordering is an interpretation that we use to eliminate the meaningless infinities that occur in field theories. So the rule goes that 'if you want to tell someone about a string of operators in quantum field theory, you have to normal order them first'. Thus, in operators like $\hat{P}$ and $\hat{H}$, in the final expression, we don't use the commutator but the normal operator operator, which will then completely subdue the issue of the appearance of Dirac-delta.

11.3 Constructing the Hilbert Space

The entire basis for this is the harmonic oscillator. In quantum SHO, we started with the vacuum and then acted creation operator successively on the vacuum to create excited states. Here, we will also do the same. Let us define the vacuum state $|\Omega\rangle$ such that,

$$\hat{a}_{\mathbf{p}} |\Omega\rangle = 0 \quad \forall \mathbf{p}$$

With such a definition, on physical grounds also, we obtain,

$$\hat{H} |\Omega\rangle = 0 \implies \text{Ground state has zero energy}$$

The vacuum can be written explicitly as:

$$|\Omega\rangle = \bigotimes_{\mathbf{p}} |\Omega\rangle_{\mathbf{p}} \quad |\Omega\rangle_{\mathbf{p}} \rightarrow \text{vac. of SHO with mom. } \mathbf{p}$$

The Hilbert space is then built as the space generated by all finite linear combinations of vectors of the form:

$$|\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\rangle \equiv \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_2}^\dagger \dots \hat{a}_{\mathbf{p}_n}^\dagger |\Omega\rangle$$

Till now, this is not exactly the Hilbert space, since we did not account for the norm of the vectors, where a huge subtlety lies.

Note that $|\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\rangle$ is an eigenstate of both $\hat{H}$ and $\hat{P}$ since,

$$\begin{aligned} \hat{H} |\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\rangle &= (\omega_{\mathbf{p}_1} + \dots + \omega_{\mathbf{p}_n}) |\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\rangle \\ \hat{P} |\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\rangle &= (\mathbf{p}_1 + \dots + \mathbf{p}_n) |\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n\rangle \end{aligned}$$

Since the creation operators commute, the multi-particle states are symmetric under exchange of particles, that is, the particles follow *Bose-Einstein* statistics.

For a general many-particle system, this ‘pre’-Hilbert space is termed as *Fock space* $\mathfrak{F}$ and is formally defined with respect to direct-sums,

$$\mathfrak{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}$$

where $\mathcal{H}^{\otimes n}$ is the n -times tensor product of the Hilbert space. In simple terms, $\mathcal{H}^{\otimes 0}$ is the space spanned by the vacuum state, $\mathcal{H}^{\otimes 1}$ is the space spanned by $|\mathbf{p}\rangle$. For $n \geq 2$, we have to consider the fermionic and bosonic properties too, that is, in our case, since these are bosons, $\mathcal{H}^{\otimes 2} = \text{Sym.}(\mathcal{H} \otimes \mathcal{H})$ is the space spanned by $|\mathbf{p}_1, \mathbf{p}_2\rangle$ such that Bose statistics is followed and so on...

Lecture 12: Making things invariant

The states that we had defined earlier, $|\mathbf{p}_1, \mathbf{p}_2, \dots\rangle$ are *improper*, in the sense that they cannot be normalised. The obvious choice, $\langle \mathbf{p} | \mathbf{p}' \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}')$ is not physical due to the presence of the Dirac-delta. We can form physical states by ‘smearing out’ these deltas, by defining:

$$|\psi\rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \psi(\mathbf{p}) |\mathbf{p}\rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \psi(\mathbf{p}) \hat{a}_{\mathbf{p}}^\dagger |\Omega\rangle$$

where $\psi(\mathbf{p})$ is an $\mathbb{L}^2$ function. However, this smearing process is not a priori Lorentz invariant. Thus, we can force the process to follow Lorentz invariance since the inception only, so that we do not run into trouble later.

12.1 Invariant Inner Product

For simplicity, consider a Lorentz boost along the x direction. Then,

$$p'^1 = \gamma(p^1 + \beta E) \quad E' = \gamma(E + \beta p^1)$$

where $\gamma = 1/\sqrt{1-(v/c)^2}$ and $\beta = v/c$ are defined as usual. Now, note that $p^\mu p_\mu = m^2 \implies E^2 = (p^0)^2 + (p^1)^2 + (p^2)^2 + m^2 \implies 0 < E = \sqrt{(p^0)^2 + (p^1)^2 + (p^2)^2 + m^2}$ under the Minkowski metric. As the transformation is only along x direction, the momentum in x and y do not change. Thus, we have:

$$\frac{\partial E}{\partial p^j} = \frac{1}{2E} \times (2p^j) \implies \frac{\partial E}{\partial p^1} = \frac{p^1}{E}$$

Now, note the transformation of the Dirac-delta:

$$\int d^3\mathbf{y} \delta^{(3)}(\mathbf{x} - \mathbf{y}) = \int d^3\mathbf{y}' \delta^{(3)}(\mathbf{x}' - \mathbf{y}') = \int \left| \frac{\partial \mathbf{y}'}{\partial \mathbf{y}} \right| d^3\mathbf{y} \delta^{(3)}(\mathbf{x}' - \mathbf{y}')$$

Comparing, we see that the Dirac-delta transforms as $\delta^{(3)}(\mathbf{x} - \mathbf{y}) = \left| \frac{\partial \mathbf{y}'}{\partial \mathbf{y}} \right| \delta^{(3)}(\mathbf{x}' - \mathbf{y}')$ where $\left| \frac{\partial \mathbf{y}'}{\partial \mathbf{y}} \right|$ is the determinant of the Jacobian matrix J of the transformation. Hence, for the Lorentz transformation, we have:

$$\begin{aligned} \delta^{(3)}(\mathbf{p} - \mathbf{k}) &= \delta(p^1 - k^1) \delta(p^2 - k^2) \delta(p^3 - k^3) \\ &= \mathfrak{J} \delta(p'^1 - k'^1) \delta(p'^2 - k'^2) \delta(p'^3 - k'^3) \end{aligned}$$

Now, let us calculate the Jacobian matrix, whose elements will be given by $J^{ij} = \frac{\partial p'^i}{\partial p^j}$ (note that since we are concerned only with 3-momentum delta function, only spatial indices will appear). Note that $\frac{\partial p'^2}{\partial p^j} = \delta_{2j}$ and $\frac{\partial p'^3}{\partial p^j} = \delta_{3j}$. Only for $j = 1$, we have:

$$\frac{\partial p'^1}{\partial p^j} = \gamma \delta_{1j} + \gamma \beta \frac{\partial E}{\partial p^j}$$

and from the dispersion relation, we have $\frac{\partial E}{\partial p^j} = \frac{p^j}{E}$. The matrix is then:

$$J \equiv \begin{pmatrix} \gamma \left(1 + \beta \frac{p^1}{E}\right) & \gamma \beta \frac{p^2}{E} & \gamma \beta \frac{p^3}{E} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies \mathfrak{J} = \det J = \gamma \left(1 + \beta \frac{p^1}{E}\right) = \frac{\gamma}{E} (E + \beta p^1) = \frac{E'}{E}$$

Thus, we obtain the transformation of the delta function as:

$$\delta^{(3)}(\mathbf{p} - \mathbf{k}) = \frac{E'}{E} \delta^{(3)}(\mathbf{p}' - \mathbf{k}') \implies E \delta^{(3)}(\mathbf{p} - \mathbf{k}) \text{ is invariant under L.T}$$

Then, we can now define the Lorentz-invariant inner product such that:

$$\langle \mathbf{p} | \mathbf{k} \rangle = (2E_{\mathbf{p}})(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}) \equiv (2\omega_{\mathbf{p}})(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k})$$

In line with this, we can therefore define a single-particle state as:

$$|\mathbf{p}\rangle = \sqrt{2\omega_{\mathbf{p}}} \hat{a}_{\mathbf{p}}^\dagger |\Omega\rangle$$

Technically, we should have made some distinct notation to differentiate between the states with and without proper normalisation, however, henceforth we will always use normalised states, so confusion might not be there.

12.2 Invariant Measure

Define the *projector* onto the single-particle state,

$$\mathbb{1} := \int \frac{d^3 \mathbf{p}}{(2\pi)^3} |\mathbf{p}\rangle \langle \mathbf{p}|$$

Here we have used the states with proper normalisation, obeying Lorentz invariance. The LHS is Lorentz invariant, since the operator essentially checks whether we have single particle or not, which is a frame independent fact. Thus, the only thing to change is the *integral measure*!

Consider the following integral,

$$\int d^4 p \delta((p^0)^2 - \mathbf{p}^2 - m^2) \Theta(p^0)$$

Note that, $d^4 p$ is Lorentz invariant. Also, from the dispersion relation, we have:

$$p^\mu p_\mu = (p^0)^2 - \mathbf{p}^2 = m^2$$

The dispersion relation is also Lorentz invariant, since it contains scalars $p^\mu p_\mu$ and m^2 . Solving for p^0 , we have $p^0 = \pm \sqrt{\mathbf{p}^2 + m^2}$. Now, the choice of which branch to choose (positive or negative) is also Lorentz

invariant, since here we are considering only *orthochronous transformations*, which does not change the time component of the momentum. Thus, overall the integral is Lorentz-invariant.

$$\begin{aligned}
 \int d^4p \, \delta((p^0)^2 - \mathbf{p}^2 - m^2) \Theta(p^0) &= \int d^4p \, \delta((p^0)^2 - E_{\mathbf{p}}^2) \Theta(p^0) \\
 &= \int d^3\mathbf{p} \int_{-\infty}^{\infty} dp^0 \, \delta((p^0)^2 - E_{\mathbf{p}}^2) \Theta(p^0) \\
 &= \int d^3\mathbf{p} \int_{-\infty}^{\infty} dp^0 \, \frac{1}{2E_{\mathbf{p}}} [\delta(p^0 + E_{\mathbf{p}}) + \delta(p^0 - E_{\mathbf{p}})] \Theta(p^0) \\
 &= \int \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}}
 \end{aligned}$$

Hence, whatever measure we were taking earlier, that is, $\frac{d^3\mathbf{p}}{(2\pi)^3}$ was not Lorentz-invariant, however, as seen above,

$$\frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \text{ is manifestly Lorentz invariant}$$

Nice! So, henceforth we will always try to write the integrals in manifestly Lorentz invariant form, for example, the field expression becomes:

$$\hat{\phi}(x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} \left(\sqrt{2\omega_{\mathbf{p}}} \hat{a}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} + \sqrt{2\omega_{\mathbf{p}}} \hat{a}_{\mathbf{p}}^{\dagger} e^{-i\mathbf{p}\cdot\mathbf{x}} \right)$$

We could technically absorb the $\sqrt{2\omega_{\mathbf{p}}}$ in the definition of the creation and annihilation operators which would make the new field expressions similar to the old ones, but it is not really needed!

Finally, we can rewrite the identity operator on one-particle states as,

$$\mathbb{1} := \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} |\mathbf{p}\rangle \langle \mathbf{p}|$$

12.3 Position states?

Consider the field operator acting on the vacuum. Since annihilation operators kill the vacuum, we only have:

$$\hat{\phi}(x) |\Omega\rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} e^{-i\mathbf{p}\cdot\mathbf{x}} |p\rangle$$

Apart from the factor $\frac{1}{2\omega_{\mathbf{p}}}$, this looks more or less like some position state since it is linear superposition of all well-defined momentum states. Also, note that:

$$\begin{aligned}
 \langle \Omega | \hat{\phi}(\mathbf{x}) | \mathbf{p}' \rangle &= \langle \Omega | \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} \left(\sqrt{2\omega_{\mathbf{p}}} \hat{a}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} + \sqrt{2\omega_{\mathbf{p}}} \hat{a}_{\mathbf{p}}^{\dagger} e^{-i\mathbf{p}\cdot\mathbf{x}} \right) \sqrt{2\omega_{\mathbf{p}}} \hat{a}_{\mathbf{p}'}^{\dagger} | \Omega \rangle \\
 &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} (e^{i\mathbf{p}\cdot\mathbf{x}}) \langle \Omega | \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}'}^{\dagger} | \Omega \rangle \\
 &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} (e^{i\mathbf{p}\cdot\mathbf{x}}) \langle \Omega | (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') + \hat{a}_{\mathbf{p}'}^{\dagger} \hat{a}_{\mathbf{p}} | \Omega \rangle \\
 &= e^{i\mathbf{p}'\cdot\mathbf{x}}
 \end{aligned}$$

We can interpret this as the position-space representation of the single-particle state $|\mathbf{p}\rangle$, exactly as in non-relativistic quantum mechanics. Thus, we have:

$$\langle \Omega | \hat{\phi}(\mathbf{x}) | \mathbf{p} \rangle = \langle x | p \rangle = e^{i\mathbf{p}\cdot\mathbf{x}}$$

Lecture 13: Heisenberg Picture

Till now, we saw that the operator $\hat{\phi}(\mathbf{x})$ has no time dependence (we defined it at $t = 0$). However, space and time should be on equal footing. Hence, it is better to define these in the Heisenberg picture, where the operators evolve in time while the states don't. Recall that a Heisenberg operator is defined from the Schrödinger operator via,

$$\mathcal{O}_H = e^{+iHt} \mathcal{O}_s e^{-iHt} \implies \frac{d\mathcal{O}_H}{dt} = i[H, \mathcal{O}_H] + \frac{\partial \mathcal{O}_H}{\partial t}$$

We consider here only free fields, that is, there is no interactions. However, in general, the Hamiltonian can be decomposed as

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int.}}$$

where $\hat{H}_0$ is the *free field*. In that case, it would be better to work in the interaction picture. However, for now we assume $\hat{H}_{\text{int.}} = 0$. Note that from Eq. 1 and Eq. 2, we can obtain the time evolution of the ladder operators as:

$$\begin{aligned} \frac{d\hat{a}_{\mathbf{p}}^\dagger}{dt} &= i\omega_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger \implies \hat{a}_{\mathbf{p}}^\dagger(t) = \hat{a}_{\mathbf{p}}^\dagger(0) e^{i\omega_{\mathbf{p}}t} \implies e^{iH_0t} \hat{a}_{\mathbf{p}}^\dagger e^{-iH_0t} = \hat{a}_{\mathbf{p}}^\dagger e^{i\omega_{\mathbf{p}}t} \\ \frac{d\hat{a}_{\mathbf{p}}}{dt} &= -i\omega_{\mathbf{p}} \hat{a}_{\mathbf{p}} \implies \hat{a}_{\mathbf{p}}(t) = \hat{a}_{\mathbf{p}}(0) e^{-i\omega_{\mathbf{p}}t} \implies e^{iH_0t} \hat{a}_{\mathbf{p}} e^{-iH_0t} = \hat{a}_{\mathbf{p}} e^{-i\omega_{\mathbf{p}}t} \end{aligned}$$

Now, we calculate the field operator in the Heisenberg picture which gives:

$$\begin{aligned} \hat{\phi}(\mathbf{x}, t) &= e^{iH_0t} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{a}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}) e^{-iH_0t} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (e^{iH_0t} \hat{a}_{\mathbf{p}}^\dagger e^{-iH_0t} e^{-i\mathbf{p}\cdot\mathbf{x}} + e^{iH_0t} \hat{a}_{\mathbf{p}} e^{-iH_0t} e^{i\mathbf{p}\cdot\mathbf{x}}) \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}}^\dagger e^{i\omega_{\mathbf{p}}t} e^{-i\mathbf{p}\cdot\mathbf{x}} + \hat{a}_{\mathbf{p}} e^{-i\omega_{\mathbf{p}}t} e^{i\mathbf{p}\cdot\mathbf{x}}) \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}}^\dagger e^{ip\cdot x} + \hat{a}_{\mathbf{p}} e^{-ip\cdot x}) \end{aligned}$$

where $p \cdot x = p^\mu x_\mu = \omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x}$ is the inner product of four momentum and four position in Minkowski space. Similarly, the conjugate momentum operator will be obtained as:

$$\hat{\pi}(\mathbf{x}, t) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\mathbf{p}}}{2}} (\hat{a}_{\mathbf{p}} e^{-ip\cdot x} - \hat{a}_{\mathbf{p}}^\dagger e^{ip\cdot x})$$

Lecture 14 & 15: Propagators and Causality

In the previous section, we saw the time-evolution of the field operator in the Heisenberg picture, essentially calculating an expression for $\hat{\phi}(\mathbf{x}, t)$. We will denote it simply as $\hat{\phi}(x)$ where x is the position four-vector while 3-vectors will be bold-faced.

Given a particle with momentum $|\mathbf{p}\rangle$, the transition to some other momentum state $|\mathbf{k}\rangle$, where $\mathbf{k} \neq \mathbf{p}$, is given by:

$$\langle \mathbf{k} | \mathbf{p} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}) = 0$$

Since the momentum states are eigenstates of the Hamiltonian, transition is not possible. However, same cannot be said for position states. To understand if transition to another state is possible, we need something measurable. The obvious choice $\langle x | y \rangle = \langle \Omega | \hat{\phi}(x) \hat{\phi}(y) | \Omega \rangle$ is not always physical, since the field operators are not Hermitian in general (eg. complex scalar field). Nevertheless, let us see what happens when we calculate this quantity.

Formally, the inner product is called the *Wightman function* and in terms of the operators, it is defined as:

$$D(x - y) := \langle x|y \rangle = \langle \Omega | \hat{\phi}(x) \hat{\phi}(y) | \Omega \rangle$$

where x and y are four-positions and $\hat{\phi}$ is the field operator in the Heisenberg picture. The Wightman function is simply the 2-point correlation function. Let us calculate an explicit expression for this:

$$\begin{aligned} D(x - y) &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \frac{1}{\sqrt{2\omega_{\mathbf{p}'}}} \langle \Omega | \{ \hat{a}_{\mathbf{p}} e^{-ip \cdot x} + \hat{a}_{\mathbf{p}}^\dagger e^{ip \cdot x} \} \{ \hat{a}_{\mathbf{p}'} e^{-ip' \cdot y} + \hat{a}_{\mathbf{p}'}^\dagger e^{ip' \cdot y} \} | \Omega \rangle \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \frac{1}{\sqrt{2\omega_{\mathbf{p}'}}} e^{-ip \cdot x} e^{ip' \cdot y} \langle \Omega | \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}'}^\dagger | \Omega \rangle \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \frac{1}{\sqrt{2\omega_{\mathbf{p}'}}} e^{-ip \cdot x} e^{ip' \cdot y} (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} e^{-ip \cdot (x - y)} \end{aligned}$$

From the above expression, it is evident that $D(x - y)$ is Lorentz invariant since the measure and integrand both are invariant. Let us consider the case for different intervals.

Case 1: $(x - y)$ is a *time-like* interval, that is, $(x - y)^2 > 0$ under Minkowski metric $(+, -, -, -)$

We can find a frame where the spatial interval is zero, that is, $\mathbf{x}' - \mathbf{y}' = 0$. Now, since $x^0 > y^0$ in our original frame and we are considering *orthochronous* Lorentz transformation, $(x')^0 > (y')^0$ too, since orthochronous transformations cannot flip the sign of the time-component. Defining $t = (x')^0 - (y')^0$, we have:

$$\begin{aligned} D(x - y) &= D(x' - y') = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} e^{-ip^0 t} \\ &= \int_0^\infty \frac{4\pi p^2 dp}{(2\pi)^3} \frac{1}{2\sqrt{p^2 + m^2}} e^{-i\sqrt{p^2 + m^2} t} \\ &= \frac{1}{(2\pi)^2} \int_0^\infty dp \frac{p^2}{\sqrt{p^2 + m^2}} e^{-i\sqrt{p^2 + m^2} t} \end{aligned}$$

Let us take $\epsilon = \sqrt{p^2 + m^2} \implies \epsilon^2 = p^2 + m^2 \implies \epsilon d\epsilon = p dp$. Substituting this in the integral, we obtain:

$$\begin{aligned} D(x - y) &= \frac{1}{(2\pi)^2} \int_0^\infty dp \frac{p^2}{\sqrt{p^2 + m^2}} e^{-i\sqrt{p^2 + m^2} t} \\ &= \frac{1}{(2\pi)^2} \int_m^\infty \frac{1}{\epsilon} e^{-i\epsilon t} (\epsilon^2 - m^2) \frac{\epsilon d\epsilon}{\sqrt{\epsilon^2 - m^2}} \\ &= \frac{1}{(2\pi)^2} \int_m^\infty e^{-i\epsilon t} \sqrt{\epsilon^2 - m^2} d\epsilon \end{aligned}$$

Now, let us take $\epsilon = m \cosh \theta \implies d\epsilon = m \sinh \theta d\theta$. Substituting this in the integral above, we obtain:

$$\begin{aligned}
 D(x-y) &= \frac{1}{(2\pi)^2} \int_0^\infty e^{-im \cosh \theta t} \sinh^2 \theta d\theta \\
 &= \frac{1}{2(2\pi)^2} \int_0^\infty e^{-im \cosh \theta t} (\cosh 2\theta - 1) d\theta \\
 &= \frac{1}{2(2\pi)^2} \left(\int_0^\infty e^{-im \cosh \theta t} \cosh 2\theta d\theta - \int_0^\infty e^{-im \cosh \theta t} d\theta \right) \\
 &= \frac{1}{2(2\pi)^2} (K_2(imt) - K_0(imt))
 \end{aligned}$$

where $K_\alpha(x) = \int_0^\infty e^{-x \cosh \theta} \cosh \alpha \theta d\theta$ is the integral representation of the *modified Bessel function of the second kind*. The integral can also be solved in the asymptotic limit, when $t \gg \frac{1}{m}$. Note that for large t and large ϵ , the phase in the exponent is large and there is rapid oscillations, leading to the overall integral being zero. Hence, the contribution to the integral will only come from around $\epsilon \approx m$.

Thus, taking $\epsilon = m + \xi, \xi \ll m$, we have:

$$D(x-y) = \frac{e^{-imt}}{(2\pi)^2} \int_0^\infty e^{-i\xi t} \sqrt{(2m + \xi)(2m - \xi)} d\xi = \frac{e^{-imt} \sqrt{2m}}{(2\pi)^2} \int_0^\infty \sqrt{\xi} e^{-i\xi t} d\xi = e^{-imt} \frac{\sqrt{2m}}{(2\pi)^2 t^{3/2}} \int_0^\infty \sqrt{z} e^{-iz} dz$$

The integral is found out to be $\frac{\Gamma(3/2)}{i^{3/2}} = \frac{\sqrt{\pi}}{2} e^{-3\pi i/4}$, by the analytic continuation of the Gamma function. Anyways, what we found is that, upto some factor,

$$D(x-y) \sim e^{-imt}$$

Case 2: $(x-y)$ is a *space-like* interval, that is, $(x-y)^2 < 0$ under the Minkowski metric $(+, -, -, -)$

We can find a frame where the temporal interval is zero, that is, $(x')^0 - (y')^0 = 0$. Let us define $\mathbf{r} = \mathbf{x} - \mathbf{y}$. Then we have:

$$D(x-y) = D(x'-y') = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} e^{i\mathbf{p} \cdot \mathbf{r}}$$

Consider $\mathbf{r}$ along the polar axis (if not, the rotate the polar axis), then we have:

$$\begin{aligned}
 D(x-y) &= \frac{1}{2(2\pi)^2} \int_0^\infty dp \int_{-1}^1 d(\cos \theta) \frac{p^2}{\sqrt{p^2 + m^2}} e^{ipr \cos \theta} \\
 &= \frac{1}{2(2\pi)^2} \frac{1}{ir} \int_0^\infty dp \frac{p}{\sqrt{p^2 + m^2}} (e^{ipr} - e^{-ipr}) \\
 &= \frac{1}{2ir(2\pi)^2} \int_{-\infty}^\infty dp \frac{pe^{ipr}}{\sqrt{p^2 + m^2}} \\
 &= \frac{\mathcal{I}}{2ir(2\pi)^2}
 \end{aligned}$$

Since square root is a multi-valued function, the integrand has branch points at $p = \pm im$ and we choose the branch cuts to be emanating from the poles and going to infinity. Let $r > 0$, so we chose the contour to close in the upper-half plane, as per Jordan's lemma. The contour, which avoids the branch cut, is shown below.

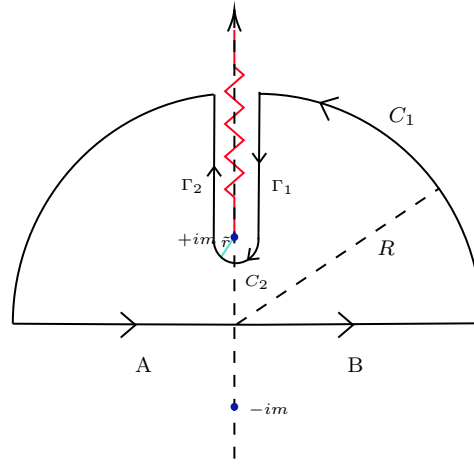
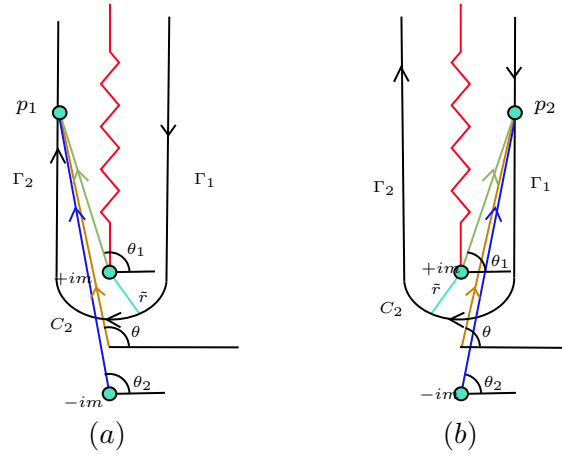


Figure 2: Contour for the integral ($r > 0$). The blue dots represent the branch points while red kinked line denotes the branch cut.

The integrals along the arcs vanish for $R \rightarrow \infty$ and $\tilde{r} \rightarrow 0$ ¹. The only contribution to the integral comes because of the discontinuity across the branch cut. Since the contour encloses no poles, then by the Residue Theorem, we have:

$$\mathcal{I} + \int_{\Gamma_1} + \int_{\Gamma_2} = 0$$

Let us now see what these two integrals along both side of the branch cut evaluates to. For that, consider the diagram below:



Note that, on the left-side of the cut, θ_1 and θ_2 both are $\pi/2$. On the right side, since $p + im$ does not encircle the branch cut, θ_2 remain $\pi/2$ however, θ_1 becomes $\pi/2 - 2\pi = -3\pi/2$, since it encircles the branch cut. We write it as following:

$$\sqrt{p^2 + m^2} = \sqrt{|p + im||p - im|} \exp\left(i \frac{\theta_1 + \theta_2}{2}\right) = \begin{cases} \sqrt{|p + im||p - im|} i & \text{right} \\ -\sqrt{|p + im||p - im|} i & \text{left} \end{cases}$$

Then, the square root becomes negative of each other on either side of the branch cut (on right, side).

¹There is a subtlety tbh; for Jordan's lemma to be applicable, the integrand other than the exponential should vanish as $R \rightarrow \infty$. However, it doesn't happen here. Yet, by some transformation with Dirac delta, we can make this thing work afaiik (Refer Padmanabhan, Problem1)!

Replacing $p = i\rho$, we have:

$$\begin{aligned} 0 &= \mathcal{I} + \int_{-\infty}^m \frac{i\rho e^{-\rho r}}{\sqrt{\rho^2 - m^2}} + \int_m^{\infty} \frac{-i\rho e^{-\rho r}}{\sqrt{\rho^2 - m^2}} \\ \Rightarrow \mathcal{I} &= 2i \int_m^{\infty} \frac{\rho e^{-\rho r}}{\sqrt{\rho^2 - m^2}} \end{aligned}$$

Substituting $\rho = m \cosh \lambda$, we have the expression:

$$\mathcal{I} = 2im \int_0^{\infty} \cosh \lambda e^{-mr \cosh \lambda} = 2im K_1(mr)$$

Therefore, the final value of the correlation function becomes:

$$D(x - y) = \frac{m}{4\pi^2 r} K_1(mr) \xrightarrow{r \rightarrow \infty} \frac{m}{4\pi^2 r} \sqrt{\frac{\pi}{2mr}} e^{-mr} \sim e^{-mr}$$

We see that for spacelike separations, the amplitude falls off as $e^{-mr} \neq 0$. This shows that there is a non-zero probability amplitude for a particle to propagate over a spacelike interval, which violates causality (as it implies faster than light travel). As we had said earlier, this quantity is not very physical. A more appropriate quantity to analyse will be the commutator of the fields, that is,

$$\begin{aligned} [\hat{\phi}(x), \hat{\phi}(y)] &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{d^3\mathbf{p}'}{(2\pi)^3} \frac{1}{2\sqrt{\omega_{\mathbf{p}}\omega_{\mathbf{p}'}}} \left\{ [\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}'}^\dagger] e^{-ip \cdot x} e^{ip' \cdot y} + [\hat{a}_{\mathbf{p}'}^\dagger, \hat{a}_{\mathbf{p}}] e^{ip \cdot x} e^{-ip' \cdot y} \right\} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} (e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)}) \\ &= D(x - y) - D(y - x) \end{aligned}$$

Thus, we define the propagator as:

$$\Delta(x - y) = D(x - y) - D(y - x)$$

In this case, for a space-like interval, we can find a transformation which will take $(x - y) \mapsto (y - x)$, that is, If a particle can travel in a spacelike direction from $x \rightarrow y$, it can just as easily travel from $y \rightarrow x$ (since there is not Lorentz invariant way to order events), as a result of which, $\Delta(x - y) = 0$, no longer causing any violation of causality. For time-like separation, we cannot find such a transformation and as a result, the commutator is non-zero.

15.1 Propagators

Note that the commutator is basically a number (with an implicit identity operator sitting quietly) and thus,

$$\langle \Omega | [\hat{\phi}(x), \hat{\phi}(y)] | \Omega \rangle = D(x - y) - D(y - x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} (e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)})$$

Let us now calculate a 4-D integral of the following form, taking suppose $x_0 > y_0$:

$$\begin{aligned} \mathcal{I} &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{dp^0}{2\pi i} \frac{e^{-ip \cdot (x-y)}}{p^2 - m^2} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \int \frac{dp^0}{2\pi i} \frac{e^{-ip^0(x^0 - y^0) + i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}}{(p^0)^2 - |\mathbf{p}|^2 - m^2} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \int \frac{dp^0}{2\pi i} \frac{e^{-ip^0(x^0 - y^0) + i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}}{(p^0)^2 - \omega_{\mathbf{p}}^2} \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \int \frac{dp^0}{2\pi i} \frac{-e^{-ip^0(x^0 - y^0)}}{[(p^0) - \omega_{\mathbf{p}}][(p^0) + \omega_{\mathbf{p}}]} \end{aligned}$$

Note that the integral has two poles at $p^0 = \pm\omega_{\mathbf{p}}$, which lies on the real line. Since we are having $x^0 > y^0$, to apply Jordan's lemma, we need to close the contour in the lower half-plane. For this case, we choose to include both the poles into the contour, hence, applying the Residue Theorem, we have:

$$\int dp^0 \frac{e^{-ip^0(x^0-y^0)}}{[(p^0) - \omega_{\mathbf{p}}][(p^0) + \omega_{\mathbf{p}}]} = -2\pi i \left(\frac{e^{-i\omega_{\mathbf{p}}(x^0-y^0)}}{2\omega_{\mathbf{p}}} + \frac{e^{i\omega_{\mathbf{p}}(x^0-y^0)}}{-2\omega_{\mathbf{p}}} \right)$$

$-2\pi i$ comes since the contour is taken clockwise. Substituting this value in the above integral, we obtain:

$$\begin{aligned} \mathcal{I} &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \left(\frac{e^{-i\omega_{\mathbf{p}}(x^0-y^0)}}{2\omega_{\mathbf{p}}} + \frac{e^{i\omega_{\mathbf{p}}(x^0-y^0)}}{-2\omega_{\mathbf{p}}} \right) \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left(\frac{e^{-i\omega_{\mathbf{p}}(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{2\omega_{\mathbf{p}}} + \frac{e^{i\omega_{\mathbf{p}}(x^0-y^0)+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}}{-2\omega_{\mathbf{p}}} \right) \\ &= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} (e^{-ip\cdot(x-y)} - e^{ip\cdot(x-y)}) \end{aligned}$$

In the second term, we had taken $\mathbf{p} \rightarrow -\mathbf{p}$. Since $\omega_{\mathbf{p}} = \omega_{-\mathbf{p}}$ and the volume element remains invariant, we obtained the second term in the final expression.

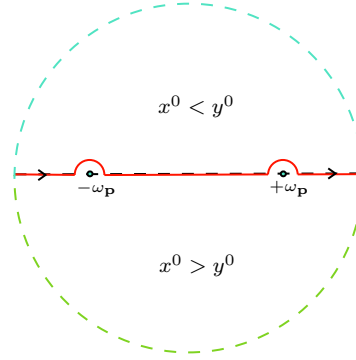


Figure 3: The contour choice for the *retarded propagator*. The green contour gives non-zero value while the blue contour gives zero since no pole is inside the contour. The red curve is common to both the contours

Thus, we see from the final expression that,

$$\langle \Omega | [\hat{\phi}(x), \hat{\phi}(y)] | \Omega \rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \int \frac{dp^0}{-2\pi i} \frac{e^{-ip\cdot(x-y)}}{(p^0)^2 - |\mathbf{p}|^2 - m^2}$$

Note that, instead of deforming the contour, we could have just shifted the poles downwards (by some amount $i\epsilon$ where $\epsilon > 0$) from the real axis, so that they fall into the contour which belongs completely to the lower half-plane. Thus, we can write the expression equivalently as:

$$\langle \Omega | [\hat{\phi}(x), \hat{\phi}(y)] | \Omega \rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \int \frac{dp^0}{-2\pi i} \frac{e^{-ip\cdot(x-y)}}{(p^0 + i\epsilon)^2 - |\mathbf{p}|^2 - m^2} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \int \frac{dp^0}{-2\pi i} \frac{e^{-ip\cdot(x-y)}}{(p^0 + i\epsilon)^2 - \omega_{\mathbf{p}}^2}$$

Note that, if we had taken $x^0 < y^0$, we would have closed the contour in the upper half-plane for Jordan's lemma to be valid and we would have got the integral as zero, since the contour did not enclose any of the poles. This is called the *Retarded Propagator* and is written compactly as,

$$D_R(x-y) = \Theta(x^0 - y^0) \langle \Omega | [\hat{\phi}(x), \hat{\phi}(y)] | \Omega \rangle$$

where $\Theta(\cdot)$ represents the Heaviside-theta function. Similar to the above, we can define the *Advanced Propagator* which vanishes for $x^0 > y^0$. In this case, the poles are shifted upwards by $i\epsilon$. We can define this as:

$$D_A(x-y) = \Theta(y^0 - x^0) \langle \Omega | [\hat{\phi}(x), \hat{\phi}(y)] | \Omega \rangle$$

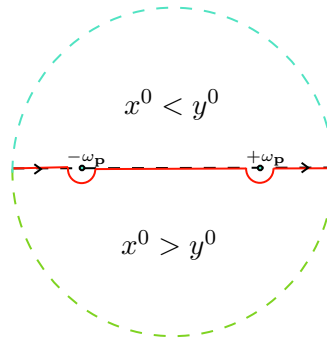


Figure 4: Contour choice for the advanced propagator.

There exists another pole prescription which gives rise to the celebrated *Feynman Propagator* which we investigate now.

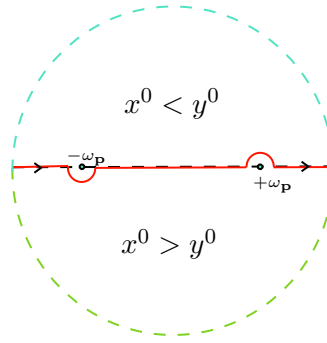


Figure 5: Pole choice for the Feynman propagator. We choose one pole to be inside and one pole to be outside the contour.

The Feynman propagator is defined as:

$$D_F(x - y) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} \int \frac{dp^0}{-2\pi i} \frac{e^{-ip \cdot (x-y)}}{(p^0)^2 - \omega_{\mathbf{p}}^2 + i\epsilon}$$

In this case, the contour is chosen such that one pole is shifted up and one pole is shifted down. Let the poles be,

$$p^0 = -\omega_{\mathbf{p}} + i\epsilon' \quad p^0 = \omega_{\mathbf{p}} - i\epsilon'$$

Then we have the following,

$$(p^0 + \omega_{\mathbf{p}} - i\epsilon')(p^0 - \omega_{\mathbf{p}} + i\epsilon') = (p^0)^2 - (\omega_{\mathbf{p}} - i\epsilon')^2 = (p^0)^2 - \omega_{\mathbf{p}}^2 + \epsilon'^2 + 2i\omega_{\mathbf{p}}\epsilon'$$

We can neglect the ϵ'^2 term compared to the other term since it's a infinitesimal quantity. Also, for the same reason, we can define $\epsilon \equiv 2\omega_{\mathbf{p}}\epsilon'$ in the integral. The Feynman propagator can then be written as,

$$\begin{aligned} D_F(x - y) &= \Theta(x^0 - y^0) \langle \Omega | \hat{\phi}(x) \hat{\phi}(y) | \Omega \rangle + \Theta(y^0 - x^0) \langle \Omega | \hat{\phi}(y) \hat{\phi}(x) | \Omega \rangle \\ &= \langle \Omega | \mathcal{T} \hat{\phi}(x) \hat{\phi}(y) | \Omega \rangle \end{aligned}$$

where $\mathcal{T}$ is the *time-ordering* operator, which brings the operator to the left with the 'greater' time, similar to the normal ordering operator encountered previously.

Feynman propagator is very important and has many applications as will be discussed later.

Lecture 16: Introducing the Dirac Equation

We saw that, when quantised, the Klein-Gordon equation provided a negative probability density, which was problematic since it had no obvious interpretation. Hence we said that, the KG equation cannot

describe a single-particle system, instead, it describes a spin-0 particle *field*. All the problems arose since, from the relativistic dispersion relation,

$$E^2 = p^2 + m^2 = p_x^2 + p_y^2 + p_z^2 + m^2 \quad (3)$$

we obtain a differential equation which is second order in both space and time.

Let us now look for something which we force to be first order in space and time from the beginning. The most sane choice is to just take the square-root of the dispersion relation, however, square-root of an operator is ill-defined and has many other subtleties. We want everything to be as simple as possible. Hence, we try to *linearise* the square-root. We thus write,

$$E = \sqrt{p^2 + m^2} = \alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3 + \beta m$$

We first promote the momentum and energy to operators, that is, $p \rightarrow -i\nabla$ and $E \rightarrow i\partial_t$. Then the Schrödinger equation becomes,

$$i\partial_t \psi = \left(-i\alpha_i \frac{\partial}{\partial x^i} + \beta m \right) \psi \quad \Longleftrightarrow \quad H\psi = E\psi$$

Since we want this to match with the actual dispersion relation Eq. 3, let us see what happens when we apply the Hamiltonian twice on the time-independent equation (which would give us an E^2).

Note that, since the *linearisation* of the square-root is extremely non-trivial, α_i and β might not be just numbers, they could be literally anything. Thus, we do not make any apriori assumptions!¹

$$H^2 \equiv \left(-\alpha_i^2 \frac{\partial^2}{\partial x^{i2}} + \beta m^2 - (\alpha_i \alpha_j + \alpha_j \alpha_i) \frac{\partial^2}{\partial x^i \partial x^j} - im(\beta \alpha_i + \alpha_i \beta) \frac{\partial}{\partial x^i} \right)$$

If the relativistic dispersion relation has to be satisfied, we find some constraints on α_i and β :

$$\begin{aligned} \alpha_i^2 &= \mathbb{1} & \alpha_i \alpha_j + \alpha_j \alpha_i &= 2\delta_{ij} \mathbb{1} \\ \beta^2 &= \mathbb{1} & \beta \alpha_i + \alpha_i \beta &= 0 \end{aligned}$$

where $\mathbb{1}$ represents the identity operator in appropriate space. Having α_i and β to be just simple scalar numbers clearly do not satisfy the equations, hence let us assume for now, that these are matrices. We then additionally impose the constraint that $\alpha_i^\dagger = \alpha_i$ and $\beta^\dagger = \beta$ since energy and momentum operator (and hence the Hamiltonian) are Hermitian. With this assumption, let us note some nice properties of these matrices.

- Let v be an eigenvector of $\alpha_i \implies \alpha_i v = \lambda v \implies \underbrace{\alpha_i^2}_{\mathbb{1}} v = \lambda^2 v \implies \lambda = \pm 1$. Similar argument goes for β . Thus, these matrices have eigenvalues either +1 or -1.
- $\alpha_i \beta = -\beta \alpha_i \implies \alpha_i \beta^2 = -\beta \alpha_i \beta \implies \alpha_i = \beta \alpha_i \beta \implies \text{Tr } \alpha_i = -\text{Tr } \alpha_i \implies \text{Tr } \alpha_i = 0$, where we have used the cyclic property of trace. Similar argument goes for β . Hence we obtain that these matrices are also traceless.
- Since the trace, that is, the sum of the eigenvalues, should equal zero, we should definitely have half of the eigenvalues as +1 and the other half to be -1. Thus, these matrices are guaranteed to be *even* dimensional.
- The relations between α_i looks suspiciously similar to the Pauli matrices and we can choose α_i to be σ_i . Alas, we run into a dangerous problem. We note that β cannot be the a two-dimensional matrix, since then it would not be consistent with the anticommutation relation with $\alpha_i \equiv \sigma_i$. Thus, the next option is to try for *four dimensional* matrices.

¹Well, we do assume that α_i and $\frac{\partial}{\partial x^i}$ commute in some sense

Let us define the following,

$$\alpha_i := \begin{bmatrix} 0 & \vdots & \sigma_i \\ \hline \sigma_i & \vdots & 0 \end{bmatrix} \quad \beta := \begin{bmatrix} \mathbb{1} & \vdots & 0 \\ \hline 0 & \vdots & -\mathbb{1} \end{bmatrix}$$

With this definition, we can indeed verify that all the conditions hold ¹ and we obtain a way to proceed further. We have an expression for the equation as,

$$i\frac{\partial\psi}{\partial t} = -i\alpha \cdot \nabla + \beta m \implies i\beta\frac{\partial\psi}{\partial t} = -i\beta\alpha \cdot \nabla + \underbrace{\beta^2}_{\mathbb{1}} m$$

Let us define the following matrices:

$$\begin{aligned} \gamma^0 &:= \beta \\ \gamma^i &:= \beta\alpha_i \end{aligned}$$

Then, using this definition in the above expression, we have:

$$i\gamma^0\partial_0\psi + i\gamma^i\partial_i\psi - m\psi = 0 \implies (i\gamma^\mu\partial_\mu - m)\psi = 0$$

Note that, we had introduced the 4-vector concept in an ad-hoc manner. However, doing this, we finally derive the well-known form of the *Dirac equation*!

Lecture 17: Dirac in Fourier Mode

We saw that in the previous section that the Dirac equation is essentially:

$$i\partial_t\psi = (-i\alpha \cdot \nabla + \beta m)\psi \quad \psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

Let the proposed solution be a plane-wave solution, since there is no interaction, and we decompose it into its Fourier modes, that is,

$$\psi(x) = e^{-iEt}\psi(\mathbf{x}) = e^{-iEt} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \psi_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{x}}$$

Substituting this in the equation above, we have:

- LHS:

$$i\partial_t\psi = i(-iE)\psi(x) = E\psi(x)$$

- RHS:

$$e^{-iEt} \int \frac{d^3\mathbf{p}}{(2\pi)^3} (-i\alpha \cdot (i\mathbf{p}) + \beta m) \psi_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} = e^{-iEt} \int \frac{d^3\mathbf{p}}{(2\pi)^3} (\alpha \cdot \mathbf{p} + \beta m) \psi_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}}$$

Equating both LHS and RHS we get an equation in $\psi_{\mathbf{p}}$,

$$\begin{aligned} 0 &= (\alpha \cdot \mathbf{p} + \beta m - E) \psi_{\mathbf{p}} \\ &= \left(\begin{pmatrix} 0 & \sigma_i p_i \\ \sigma_i p_i & 0 \end{pmatrix} + \begin{pmatrix} (m-E)\mathbb{1} & 0 \\ 0 & -(E+m)\mathbb{1} \end{pmatrix} \right) \psi_{\mathbf{p}} \\ &= \begin{pmatrix} (m-E)\mathbb{1} & \sigma \cdot \mathbf{p} \\ \sigma \cdot \mathbf{p} & -(m+E)\mathbb{1} \end{pmatrix} \psi_{\mathbf{p}} \end{aligned}$$

¹Note that these matrices are 4×4 matrices. Each element in the matrix are indeed 2×2 matrices.

Now, for a solution to exist, the matrix should be *singular*, that is,

$$\det \begin{pmatrix} (m-E)\mathbb{1} & \sigma \cdot \mathbf{p} \\ \sigma \cdot \mathbf{p} & -(m+E)\mathbb{1} \end{pmatrix} = 0$$

Note that, we can calculate the determinant in the “usual” sense for block-matrices also, if the elements in the first column commute. Since first element is the identity matrix, it commutes with $\sigma \cdot \mathbf{p}$ and hence, we can write:

$$\begin{aligned} 0 &= \det ((E^2 - m^2)\mathbb{1} - (\sigma \cdot \mathbf{p})(\sigma \cdot \mathbf{p})) \\ &= \det ((E^2 - m^2)\mathbb{1} - (\sigma_i p_i)(\sigma_j p_j)) \\ &= \det ((E^2 - m^2)\mathbb{1} - (\sigma_i \sigma_j) p_i p_j) \\ &= \det \left((E^2 - m^2)\mathbb{1} - |\mathbf{p}|^2 + \sum_{i < j} (\sigma_i \sigma_j + \sigma_j \sigma_i) p_i p_j \right) \\ &= \det ((E^2 - m^2 - |\mathbf{p}|^2)\mathbb{1}) \end{aligned}$$

From this, we finally obtain that $E^2 - m^2 - |\mathbf{p}|^2 = 0 \implies E = \pm \sqrt{m^2 + |\mathbf{p}|^2}$. We can see that, the equation says that for each momentum $\mathbf{p}$, we have two positive-energy solutions and two negative-energy solutions. Now let us find the eigenvectors. Assume it to be of the form,

$$\psi_{\mathbf{p}} = \begin{pmatrix} \phi \\ \xi \end{pmatrix}$$

Then, substituting this in the eigenvalue equation we obtain two equations:

$$\begin{aligned} 0 &= (m-E)\phi + (\sigma \cdot \mathbf{p})\xi \\ 0 &= (\sigma \cdot \mathbf{p})\phi - (m+E)\xi \end{aligned}$$

Solving for ϕ and ξ , we get $\xi = \frac{(\sigma \cdot \mathbf{p})\phi}{m+E}$ and $\phi = \frac{(\sigma \cdot \mathbf{p})\xi}{E-m}$. Now, note that the other equations gives a trivial result and thus, ϕ can be any arbitrary vector. Now, let us first take the particle at rest, that is, $\mathbf{p} = 0$. Then the eigenvalues are just $\pm m$ and the equation reduces to:

$$\begin{pmatrix} m & 0 \\ 0 & -m \end{pmatrix} \begin{pmatrix} \phi \\ \xi \end{pmatrix} = \pm m \begin{pmatrix} \phi \\ \xi \end{pmatrix}$$

Since the matrix is diagonal, the eigenvectors are trivially,

$$(\psi_{\mathbf{p}})_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (\psi_{\mathbf{p}})_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (\psi_{\mathbf{p}})_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad (\psi_{\mathbf{p}})_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

We can check that, the first two correspond to the positive energy solutions and the other two correspond to the negative energy solutions. Now, let us come back to $\mathbf{p} \neq 0$. The matrix $\sigma \cdot \mathbf{p}$ comes in the expression quite often and calculating that, we find that:

$$\sigma \cdot \mathbf{p} \equiv \begin{pmatrix} p_3 & p_1 - ip_2 \\ p_1 + ip_2 & -p_3 \end{pmatrix}$$

Now, let us take a vector v_m whose m^{th} position has 1 and all other elements are zero.

$$(Mv_m)_i = M_{ij}(v_m)_j = M_{ij}\delta_{mj} = M_{im}$$

- CASE 1: $\phi = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \implies \xi = \frac{1}{E+m} \begin{pmatrix} p_3 \\ p_1 + ip_2 \end{pmatrix}$

- CASE 2: $\phi = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \xi = \frac{1}{E+m} \begin{pmatrix} p_1 - ip_2 \\ -p_3 \end{pmatrix}$
- CASE 3: $\xi = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \phi = \frac{1}{E-m} \begin{pmatrix} p_3 \\ p_1 + ip_2 \end{pmatrix}$
- CASE 4: $\xi = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \phi = \frac{1}{E-m} \begin{pmatrix} p_1 - ip_2 \\ -p_3 \end{pmatrix}$

Thus, the final solutions for non-zero momentum are:

$$(\psi_{\mathbf{p}})_1 = \mathcal{N}_1 \begin{pmatrix} 1 \\ 0 \\ \frac{p_3}{E+m} \\ \frac{p_1 + ip_2}{E+m} \end{pmatrix} \quad (\psi_{\mathbf{p}})_2 = \mathcal{N}_2 \begin{pmatrix} 0 \\ 1 \\ \frac{p_1 - ip_2}{E+m} \\ \frac{-p_3}{E+m} \end{pmatrix} \quad (\psi_{\mathbf{p}})_3 = \mathcal{N}_3 \begin{pmatrix} \frac{p_3}{E-m} \\ \frac{p_1 + ip_2}{E-m} \\ 1 \\ 0 \end{pmatrix} \quad (\psi_{\mathbf{p}})_4 = \mathcal{N}_4 \begin{pmatrix} \frac{p_1 - ip_2}{E-m} \\ \frac{-p_3}{E-m} \\ 0 \\ 1 \end{pmatrix}$$

where $\mathcal{N}_i$ are some normalisation constants. Note that, if $\mathbf{p} = 0$ then the first two solutions correspond to the positive energy while the other two correspond to the negative energy. And thus, we define these solutions corresponding the $E = \sqrt{m^2 + |\mathbf{p}|^2}$ and $E = -\sqrt{m^2 + |\mathbf{p}|^2}$ respectively.

The initial interpretation of the negative energy solutions were that the ground state (vacuum) was such that all the negative energy states were filled. Excitations happened such that a particle with negative energy occupied a positive energy state, leaving a 'hole' in the negative energy domain, which represented an *anti-particle*.